



**NATIONAL UNIVERSITY**  
of Computer & Emerging Sciences

# **Fundamentals of Software Engineering**

## **Final Project Report**

VeriMed: Pharmaceutical Supply Chain & Verification System

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## Chapter 1: Introduction

### 1.1 Project Background

The pharmaceutical global supply chain could be described as the most complex of all supply chains. The objective of this supply chain is to see to it that the medicines are effectively supplied to the [customer](#). It is because number of individuals involved in the process, such as producers, distributors,

logistics firms, drug stores and hospitals. Although that this approach enables pharmaceutical companies access to a global market, still bad communication is its major flaw. Among these issues, the alarming one is the fake medicines. It no longer seems impossible to counter not only expensive medications but nowadays even important medications like antibiotics are prone to being counterfeit products. As per the statistics in reports made by WHO counterfeit medicines make a significant proportion of the entire pharmaceutical industry. The problem gives rise to treatment failures, health crises, antibiotic resistance, and deaths. It can also cause more damage than any social or human loss. If fake medicine makers copy the packaging of a known company, people will stop trusting that brand. Once the trust is lost, it becomes very difficult to get it back. In past the tracking such products was done by local computer system or by writing records by hand. For example, a pharmacy writes down the medicines got from a wholesaler. But there was no way to confirm if the medicine is real or not.

VeriMed is designed to solve this problem. VeriMed is a modern online system that can work on the cloud. It makes the pharmaceutical supply chain clear, secure, and trustworthy. Instead of using separate and disconnected databases, VeriMed uses one digital system. verimed keeps a complete record of every step, so the medicine can be tracked safely . Product tracking system is not hidden. It is open and transparent. Companies can manage their stock , and people can confirm that the medicine is safe or not.

## **1.2 Problem Statement**

At the very core of the issue, that was encountered during tracking of pharmaceutical products was the inherent problem with the already implemented design of the system itself, which employs disparate data systems. Right now, the process entails the use of completely independent databases and it operated by different entities, wherever important data is required for tracking the product is not shared throughout the system. When the medication leaves the original producer, it becomes very difficult to track the product's true source. There will always be "blind spots" in the supply chain in such a setting. This is because the counterfeiters exploit these very blind spots in order to bring fake medication into the supply chain, at a point where the medication is transferred from the secondary wholesalers to the local pharmacies. Because most of the pharmacies are using either paper invoices that can be easily forged, or POS systems which can be manipulated, they cannot verify the medications that are delivered to them.

Moreover, the faulty system renders it impossible for legitimate manufacturers to monitor the entire lifespan of their products; as a result, it becomes unfeasible to recall their products selectively. Thirdly, it makes the weakest link in the chain – the consumer – vulnerable since there is no means by which the consumer can verify whether the drugs he/she buys are authentic.

## **1.3 Objectives**



The primary objective of the VeriMed project is to build and develop a unique software application that would solve the problem of data silos in the pharmaceutical sector. This main objective is further split into several sub-objectives, which are:

- **Building and designing a role-based web interface:** The web interface would be expected to bring together the three most important entities, that is, the pharmaceutical manufacturers, retail pharmacy practitioners, and end users, under one database architecture.
- **Development of a cryptographic QR-code tracking mechanism:** The platform would automatically generate a unique and hash encrypted QR code for each newly manufactured batch of medicines. These codes will act as the virtual passport for the physical medicine, documenting its route from its production to its ultimate distribution.
- **For inventory management of medicines:** The automation system should simplify things for the retail pharmacies, helping them deliver their products through QR codes and manage their inventory according to the FEFO rule. Through this, there is absolutely no possibility of the retail pharmacies distributing their expired medicines and incurring additional expenses.

- **For making the information accessible and providing consumers with some power :** It is required to design a consumer friendly and scanning-enabled portal for the VeriMed system. This would help the end users to confirm the validity, origin, and expiry date of their medicines using their ordinary mobile phones without registering themselves beforehand.

## 1.4 Scope of the Project

first process involves knowing the exact scope of the VeriMed System in order to create a effective and better software application. So in general, the scope of the system has the full creation of the role-based encryption system that will consist of three related web applications.

The first phase of the project involves the creation of the **Manufacturer Dashboard** system. This is where the data life cycle begins. Using the software, the drug manufacturers will log into their account, add new batches of drugs, input the chemical composition, set the important manufacturing and expiry dates, and create QR codes for each batch of medicines.

Second, the scope includes the **Retailer Portal**. This module is designed specifically for the workflow of local pharmacists. It includes the integration of the scanner interface where the retailers will scan the physically delivered shipments and validate the sources of the shipment right away from the central MySQL database, receiving the products electronically. In addition, it includes creating an inventory management dashboard that automatically sorts out the locally available products using FEFO principles and informs the pharmacist regarding those products that are about to expire soon.

Thirdly, the scope includes the creation of the **Patient Interface**. The Patient Interface is a simple and mobile-friendly web interface where no registration from the user's end is required. Any consumer can scan the QR code on the product through their device's camera to get the validation whether the product is authentic or not in a visual form along with the provenance of the product.

**Exclusions from Scope:** To make this project more feasible and confined to a narrow scope some of the features were deliberately excluded such as Physical Shipping logistics, GPS tracking, Hardware integration (monitoring storage of medicines), this makes Verimed more focused on the software solution.

## 1.5 Overview of the System

The design of VeriMed uses a common client-server system. The backend of the system is done by Python and the Flask web framework. We chose Flask as it is simple, has good routing features, and works well with security tools used for QR.

The backend work as a bridge between system and database. Database is organized in 3rd Normal Form, which means the data is stored in a clean and efficient way.

The frontend is in HTML5, CSS3, and Bootstrap 5. Pharmacists can use tablets to manage

medicines and patients can use their smartphones to scan QR codes. The system is mobile-friendly, so it works well on phones and is easy to use.

The system works by changing the status of product in steps. First, the manufacturer creates the product and marks it as “Active / Produced.” When different people scan the QR code, such as retailers or customers, the system updates the product’s status in the database. Each update is permanently saved, so it cannot be changed later. It helps prove that the information given to the customer is real.

## **Chapter 2: System Analysis and Requirements**

### **2.1 Existing System**

Existing systems involve mainly the basic point of sale system and the inventory management systems. This system has the capability to keep track of the local stock, but lacks the link with the manufacturer's database. The absence of this link makes the tracing of the drugs and its source very difficult when the shipment arrives and uses the invoice method that could be easily bypassed.

### **2.2 Proposed System**

The system generates a distinct, secure QR code issued when a batch is manufactured. Once the shipment actually arrives, the retailer then scans that exact same barcode into the database to indicate nothing has been changed in custody of that item. To conclude, the consumer scans the very same barcode to open product history.

### **2.3 Stakeholders Identification**

- Manufacturers of Drugs: They are themselves responsible for producing the drug.
- Drug Wholesalers & Retailers: Those responsible for marketing and distribution of the drug.
- Consumers of Drugs: Those who use the drug for the intended purpose.
- IT Managers: Individuals responsible for management of VeriMed system.

### **2.4 Functional Requirements**

- FR-01: The system shall enable authorized Manufacturers to enter details of formula to create a production batch.
- FR-02: The system shall automatically create a unique QR code associated with Batch ID.
- FR-03: The system shall enable Retailers to scan the QR codes and update the inventory automatically through FEFO.
- FR-04: system has to enable users to scan QR codes to check for authenticity without logging in.

## 2.5 Non-Functional Requirements

### 1. Security

- **Data Protection:** All supply chain data is appropriate for your system in such a manner that all sensitive information must remain confidential and the integrity of such data. No passwords should ever be stored in a database without being cryptographically hashed first, so that access to the database does not grant access to user accounts.
- **Ensuring Anti Forgery :** Hash encoding for the QR codes generated for the respective batches of medicine must be secured in random manner such that they cannot be predicted, copied or tampered with by any counterfeit manufacturer.
- **Role Based Access Control:** The Taskmaster must enforce rigid access boundaries, only a validated Manufacturer can create batches and input quantities, likewise none but an authenticated Retailer has permission to update Inventory States.

### 2. Performance and Responsiveness

- **Query Latency:** The query must be very low latency to make real time feasible. In particular, the /verify endpoint for public-facing QR code verification queries must process and respond with authenticity results back to the end-user in less than 3 seconds.
- **Database Efficiency:** The read/write operations of the Fefo-trigger service and its associated data leading to the Retailer dashboard must operate with extremely low latency in order to accommodate high-throughput pharmacy environments; This means that the QR codes produced for the medicine batches should use secure, randomised hashing techniques so they cannot easily be predicted, cloned or modified by counterfeiters.

### 3. Usability and Accessibility

- **Mobile-First Design:** As retail pharmacists and public consumers will mostly access the system through handheld devices then all frontend interfaces (HTML5/Bootstrap 5) must be responsive, mobile screen optimized. All supply chain data must be supported with security controls to maintain confidentiality and integrity. Never store user passwords in plain text in the database, always hash them cryptographically before storage making it much less likely to gain access.
- **Intuitive Workflow:** The verification scanner must require no training and no registration of an account to be used, so the verifying the medicines is easy and fast.

### 4. Reliability and Availability

- **System Uptime:** to support many users accessing, scanning, entering data at the same time the architecture should be capable.
- **The importance of Error Handling:** Sending out an appropriate error message for example when the scan fails meaning full message should be printed that is understandable by the user.

## 5. Data Integrity and Maintainability

- **Database Normalization:** The database is written in mySql language in 3NF, This helps to avoid redundancy, update anomalies.
- **Framework Standards:** the standard to implement the backend is that it should be written in such a way that the system becomes flexible for the future for example ensuring scaling, maintainability and ensures a good structure.
- **Role Based Access Control:** There should be some control over the functionalities which can be performed by a certain role only for example the user logged in with the role of manufacturer can generate batches.

## 2.6 Feasibility Analysis

1. **Technical Feasibility:** As we chose tools like Python, Flask, my sql and react, they are easy to integrate hence helping make secure codes and good data base connection.
2. **Economic Feasibility:** Since this was an initial phase we did not need any paid tools, hardware or cloud storage we made use of free tools to achieve this project.
3. **Operational Feasibility:** Since the website will be highly responsive, with easy User Interface and big buttons, fit for any age group, no excessive computer skills needed thus this website will be easy to use.

# Chapter 3: System Design

## 3.1 System Architecture

VeriMed has a Layered + Integrated Client Server model. This architecture was selected in order to make sure that UI is responsive, with the backend handling the QR code and validation logics for medicines while tracking.

### 3.1.1 Overall System: Layered Architecture

The layered Architecture helps in separating modules, which makes the system easy to manage and more flexible

- **Layer 1 Frontend:** In order to develop UI, languages such as React and HTML can be used, since they are universal languages, they help make the system responsive for every possible device, such as phones or pc's.
- **Layer 2 Backend:** for backend python was chosen with flask due to its easy access to libraries which helped a lot to integrate QR codes, this model is light and fast.
- **Layer 3 Database:** developed using mysql in 3nf form, helps increase data integrity and reduce anomalies.

**Justification:**

- **Modular Structure:** Decoupling the frontend, from the backend is really helpful. It allows the team to refine the system without messing with parts of the system that are being used. This way the team can work on the frontend. The backend can keep doing its own thing without any issues. The team can make changes to the system without interfering with logics being used in the implementation of the system.
- **Secure Scalability:** The application layer is like a bridge to the MySQL database. It helps keep the persistence layer from the frontend. This means the frontend does not get to access the persistence layer. The application layer acts as a bridge, to the relational MySQL database.

### **3.1.2 Backend Communication: Client-Server Architecture**

From a technical perspective, the project is structured using the classical architecture of Client Server where the communication protocol employed was RESTful API.

- **Client:** The browser of the user makes requests for performing specific functions like creating a new form of batch creation or scanning the QR Code.
- **Server:** The Flask application written using Python acts as the medium between the client and the MySQL database.

**Justification:**

- **Efficient Data Exchange:** Using this architecture guarantees the efficiency of data exchange, which directly contributes to the achievement of the objective of resolving all queries from QR codes within 3 seconds.
- **Secure Operation Isolation:** Such architecture will ensure isolation of operations requiring protection, such as encryption and calculation of FEFO sequence, from other functions that take place within the application.

### 3.1.3 Internal Logic: MVC

In order to structure the backend code, an internal routing is done based on the MVC architecture pattern.

- **Model:** Defines the structures for the data which are linked to the tables within the MySQL database (users, batches, inventory).
- **View:** the view consists of the highly responsive HTML pages which are based on Bootstrap 5 that are visible to the enduser for example Manufacturer dashboard and Retailer portal.
- **Controller:** The controller consists of the routes from the Flask API that receive user inputs, validates the inputs using logic, works with the Model layer, and informs the View what to do next.

#### Justification:

- **Workstream Separation:** The separation of the View from the Model makes it easier for the team to enhance the UI without affecting the backend Python code's architecture.
- **Request Centralization:** Only the Controller can accept user requests. This component verifies user input data such as manufacturing dates and formula details before passing them to the database.

### 3.1.4 Data Integration: Data Pipeline

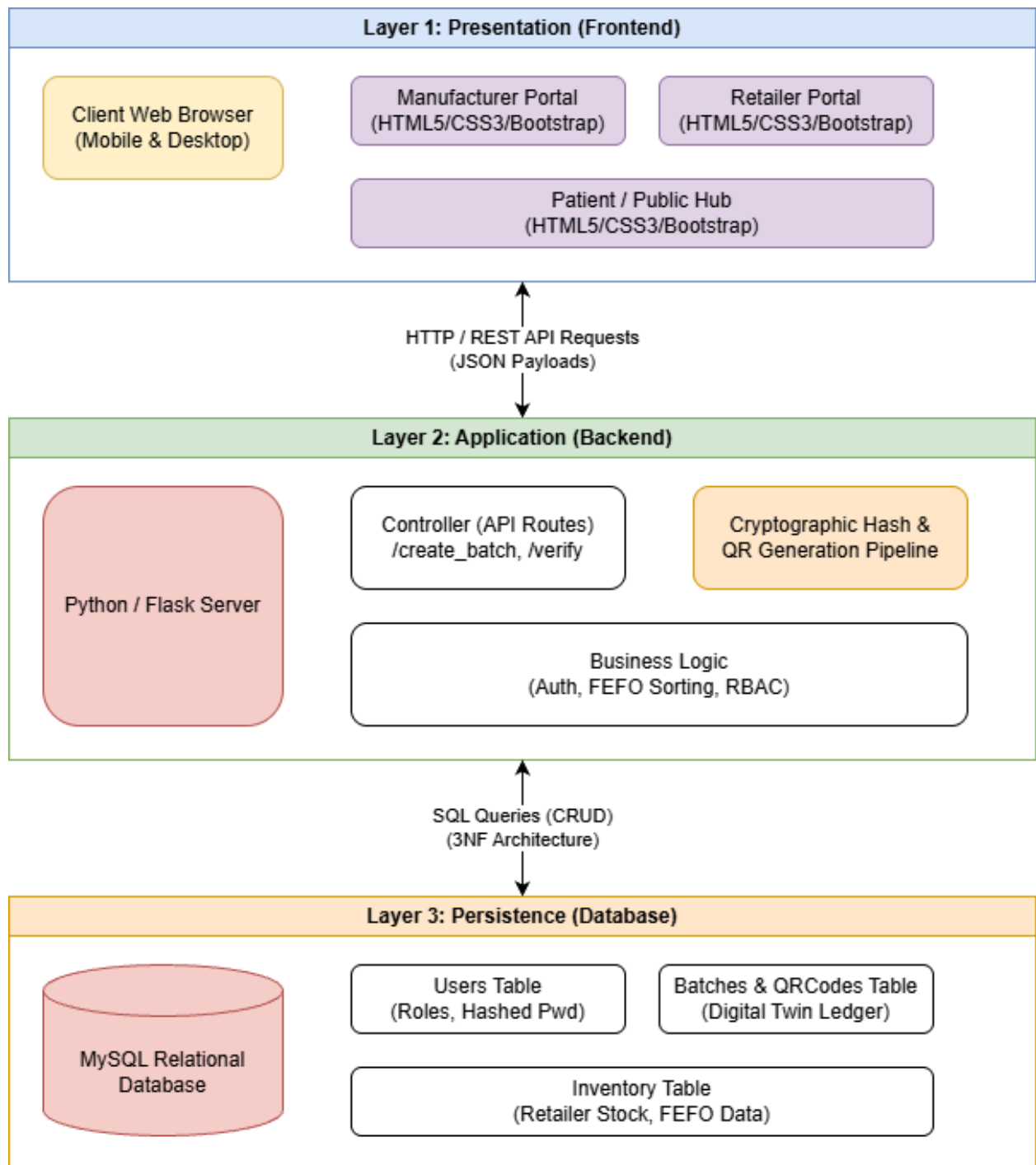
- **Step 1 Trigger:** The Manufacturer inputs formula details and dates to generate a production batch via the web form.
- **Step 2 Hash Generation:** The backend makes a unique, safe, and hash encrypted QR code that is linked to the Batch ID.
- **Step 3 Storage:** The MySQL database safely stores the generated batch record and its corresponding cryptographic hash.
- **Step 4 Validation:** after the patient/retail store owner has scanned the QR code the user will send the hash value to the main server, which will then confirms that whether there is a match by checking the database and if yes it returns the history of the product.

#### Justification:

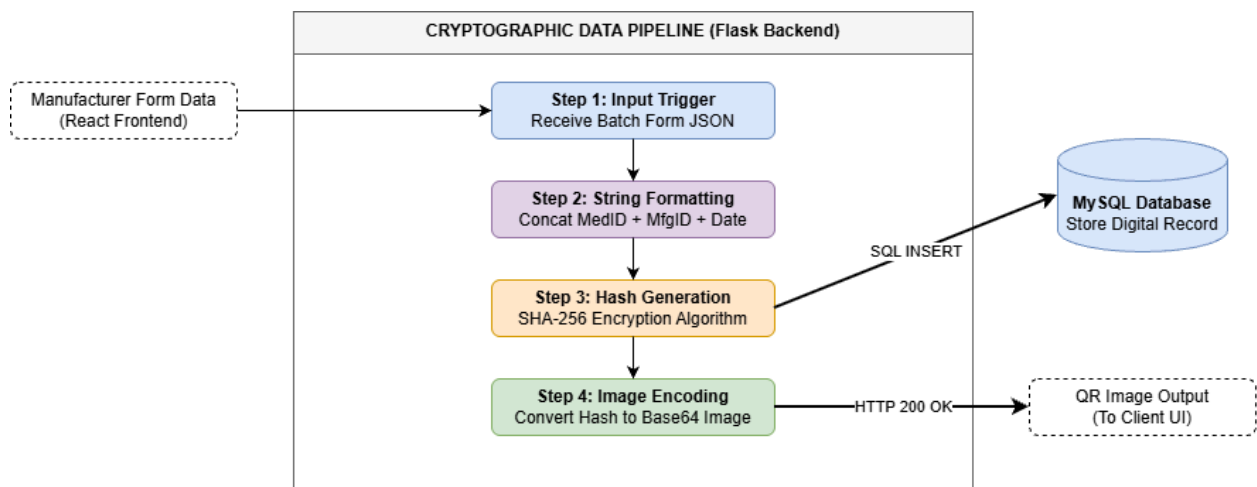
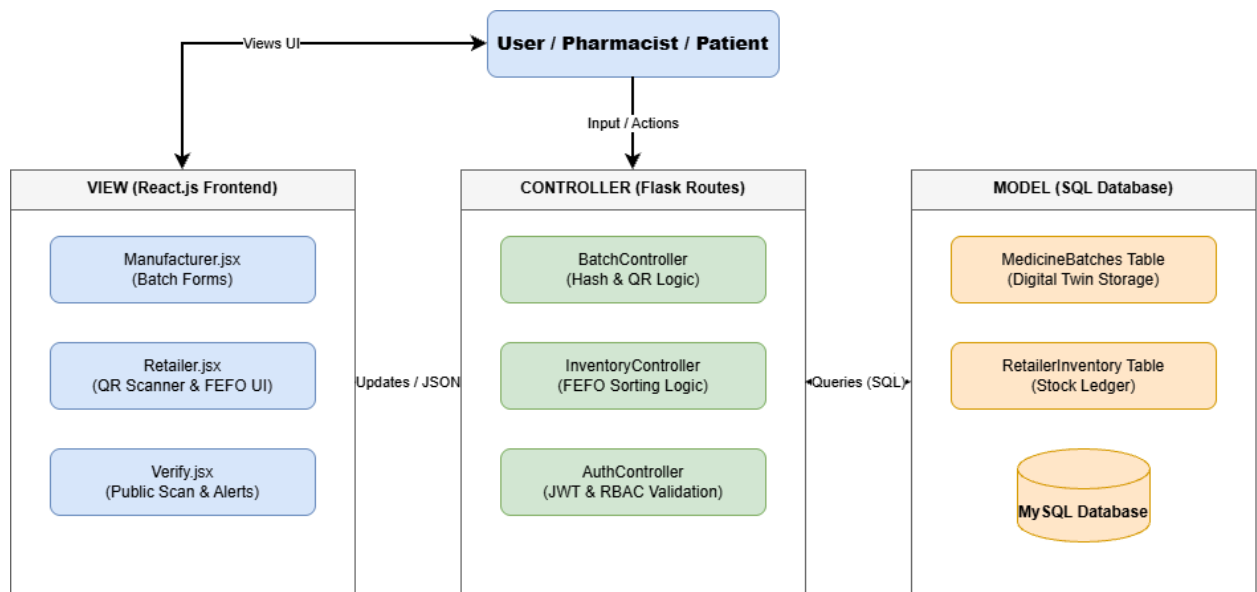
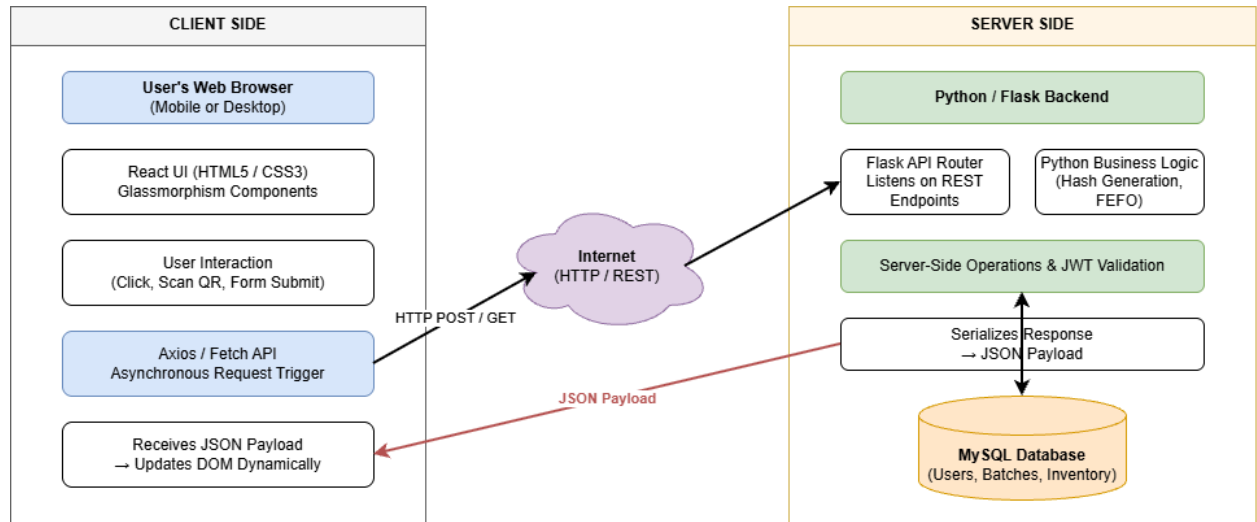
- **System Reliability:** putting these process through a strict pipeline makes sure that the information provided to the end-user is correct and prevents any kind of false data from entering the chain.

### **3.1.5 Architecture Diagram Components**





**Figure 3-A: Layer Diagram**



**Figure 3-B: Architecture Diagram**

### 3.2 Use Case Diagram



### 3.3 Use Case Descriptions

#### Use Case 1: Generate New Batch

1. **Actor:** Manufacturer
2. **Purpose:** Creation of the electronic record for a newly produced batch of medicine and generation of its QR code.
3. **Precondition:** Manufacturer is signed in to the manufacturer portal.
4. **Process:**
  - Manufacturer goes to create batch section
  - The Manufacturer enters all the necessary details about the newly created batch, for example medicine name, composition.
  - The Manufacturer clicks "Generate & Create Batch".
  - The system validates the entered data.
  - The system calculates unique hash value, to serve as the identifier of the batch.
  - The system stores the created batch in the MySQL database.
  - The system generates a visual QR code that can be downloaded by the user, which contains an encrypted Batch ID.
5. **Postconditions:** A new batch has been successfully created and is waiting to be dispatched/scanable by the retailer.

#### Use Case 2: Scan Incoming Delivery

- **Actor:** Retailer (Pharmacist)
- **Goal:** the goal is to digitally receive delivery of medicine batch from the manufacturer and then add it to the local pharmacy/retailer's inventory (automatically).
- **Pre-requisites:** before scanning the delivery the Retailer should be logged into the Retailer Portal and should have access to a device camera in order to scan the QR code.
- **Main Scenario:**
  - The Retailer taps on the "Receive Delivery" button from his/her dashboard.
  - Camera access request is prompted; scanner is displayed.
  - The Retailer uses the camera to scan the QR code on the delivery package of the medicine.
  - The QR hash is decoded by the system, which further queries the central database.
  - The system verifies that it is an authorized delivery and fetches batch information (batch quantity, expiry date).
  - The Retailer confirms quantity received and taps on the "Add to Inventory."

- The system updates the database and marks the batch status as “At Retailer.”
- The system updates the Retailer’s inventory dashboard automatically, where it sorts out the medicines according to FEFO logic.
- Alternate Scenario: if the verification fails(for example wrong credentials), a warning message will display and then the retailer will not be able to add batch to the medicine to the inventory.

### Use Case 3: Verify Authenticity (Public)

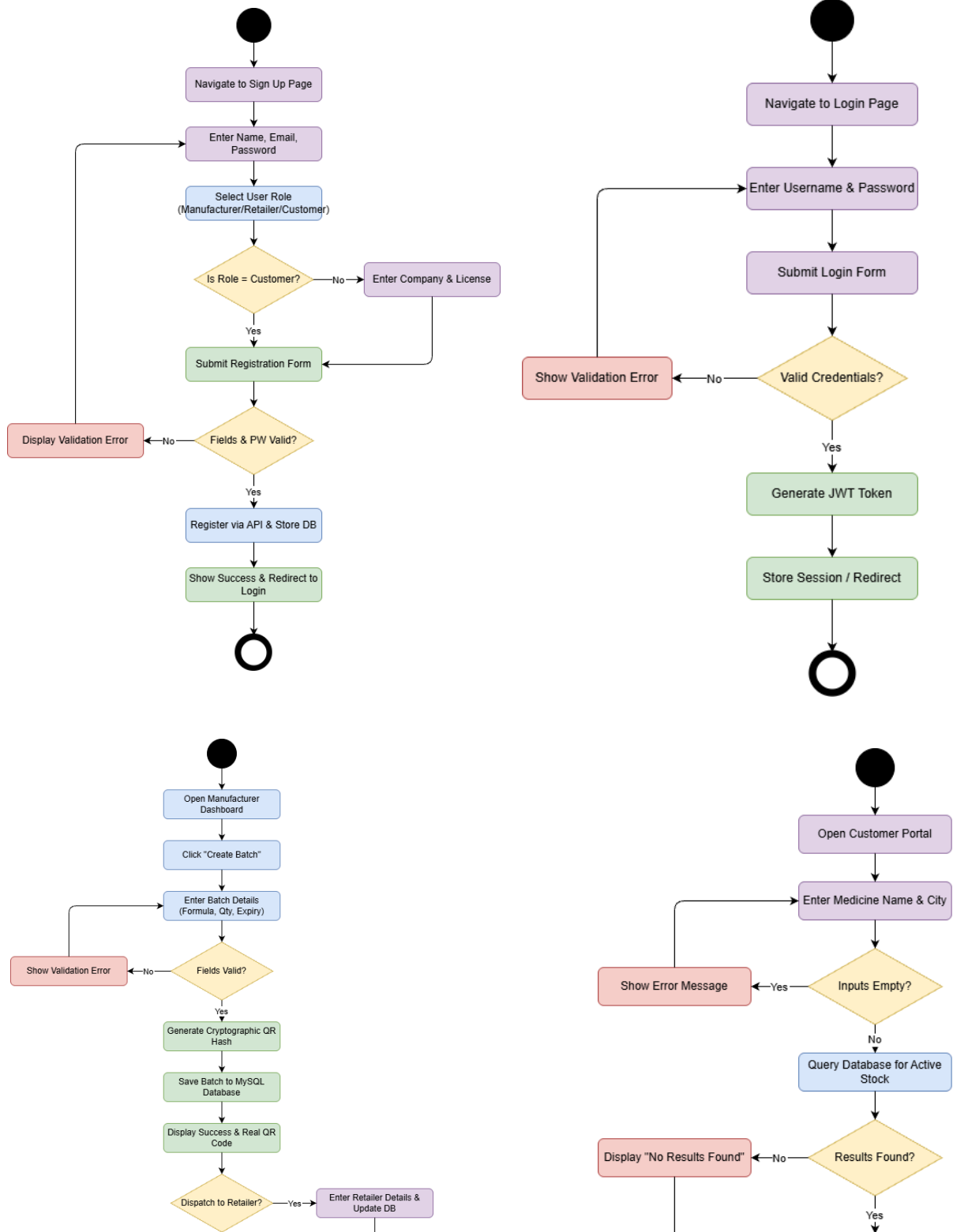
- **Actor:** Patient / Customer
- **Objective:** To confirm if the medicine they bought is real and safe to ingest without having to create a personal account.
- **Precondition:** The Patient should have a smart phone or computer with Internet connection.
- **Primary Flow of Events:**
  - The Patient accesses the VeriMed official website and goes to the "Verify Your Medicine" section.
  - The Patient decides to scan the medicine via camera or type in the ID number found on the packaging.
  - The Patient either scans the QR Code or types the ID.
  - The system will automatically search the database for a match between hash/ID.
  - The system provides a visual outcome (Green for Real, Red for Fake).
  - If confirmed real, the system will show the complete provenance of the product such as its name, production date, and expiration.
- **Postconditions:** The Patient is provided with the knowledge regarding his medicine's authenticity and safety.

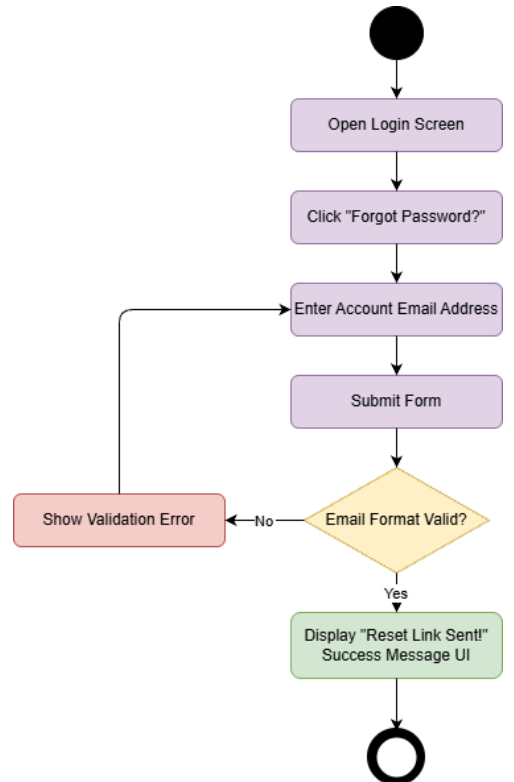
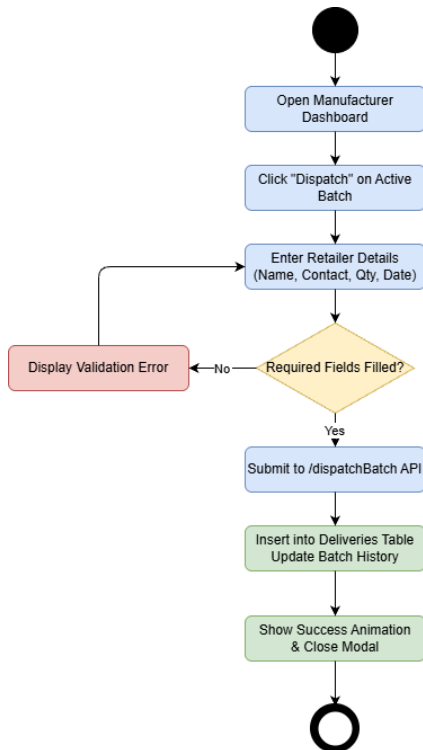
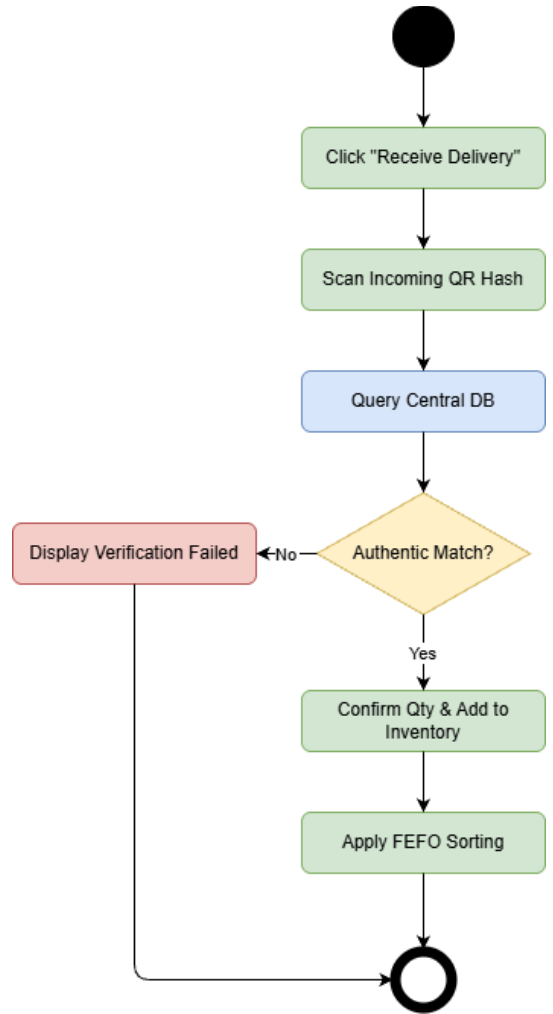
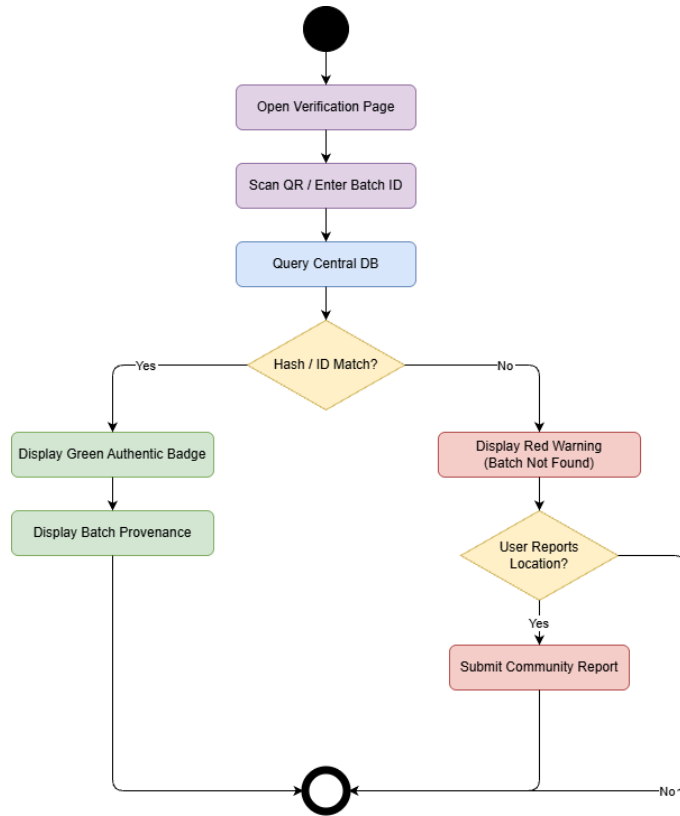
### Use Case 4: Report Suspicious Location

- Actor: Patient / Customer
- Purpose: To warn the community as well as the concerned authority about fake drugs being sold from a particular pharmacy.
- Prerequisites: The Patient is notified by the system that their scanned drug was found to be “Batch Not Found” / counterfeit.
- Primary Flow:
- After the Patient has undergone an authenticity check for the product, he is informed through a pop-up message containing “Report This Location” button.
- The Patient presses the button in order to fill up the reporting form.
- Patient fills out the field regarding the location where he purchased the drug ("Raja Bazaar, Rawalpindi") and adds a description (optional).
- The patient presses the submit button.

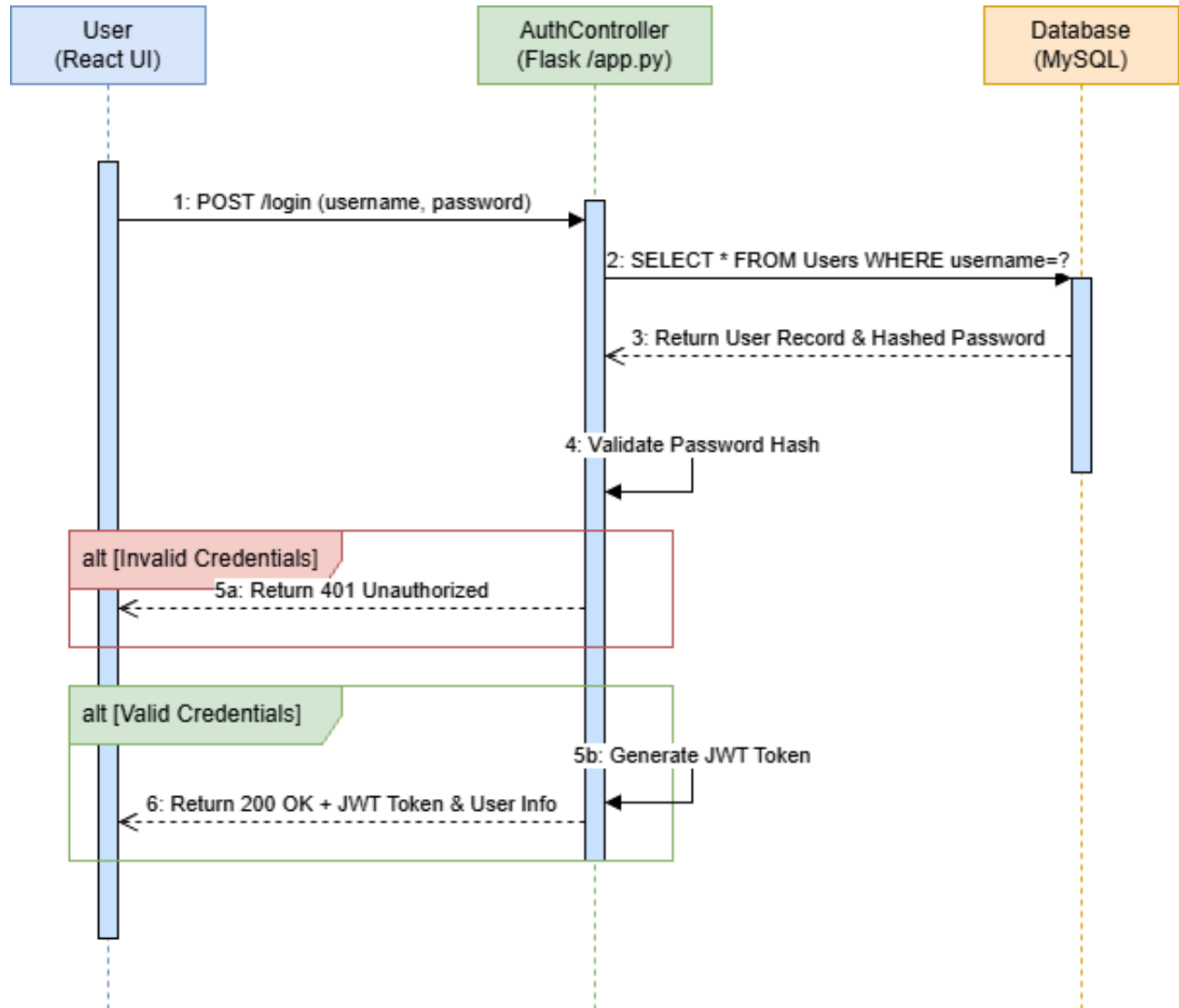
- The system adds the location to the database under “Under Review” status.
- A message of confirmation is displayed and a link is provided to see community reported locations map.
- Postconditions: The location is added to the database and marked on the “Fake Drugs Location” board.

### 3.4 Activity Diagrams

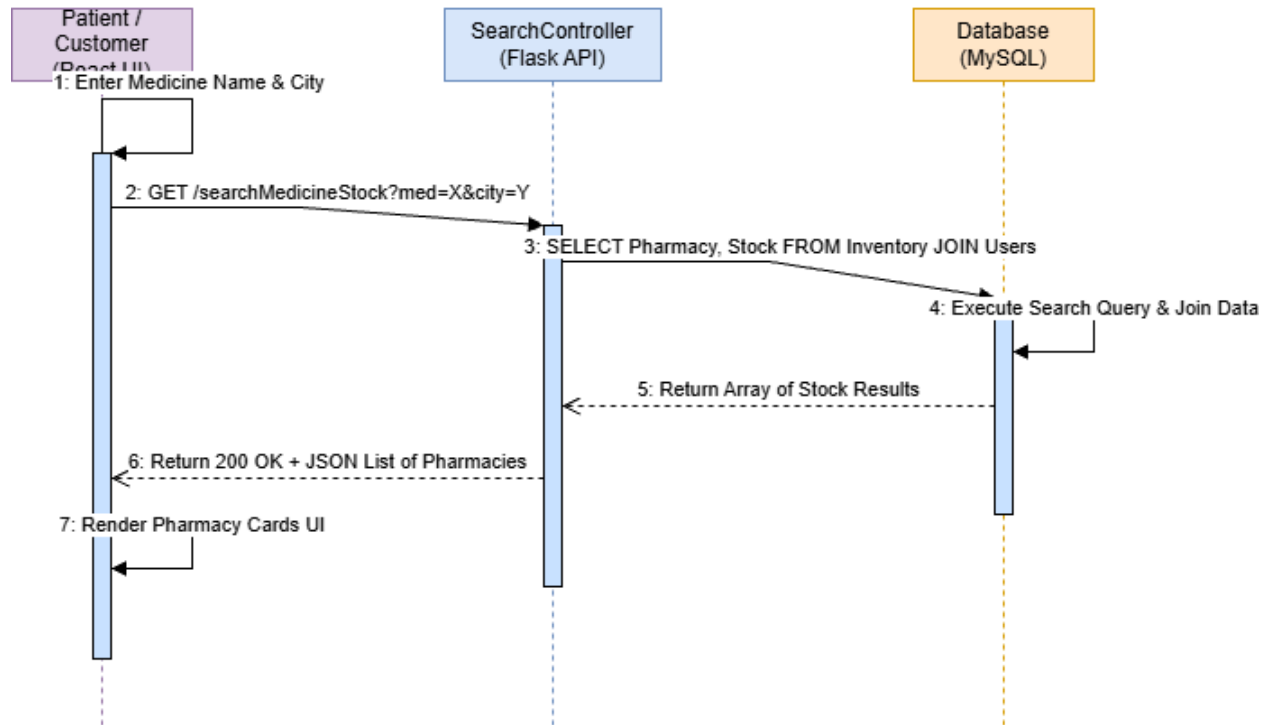


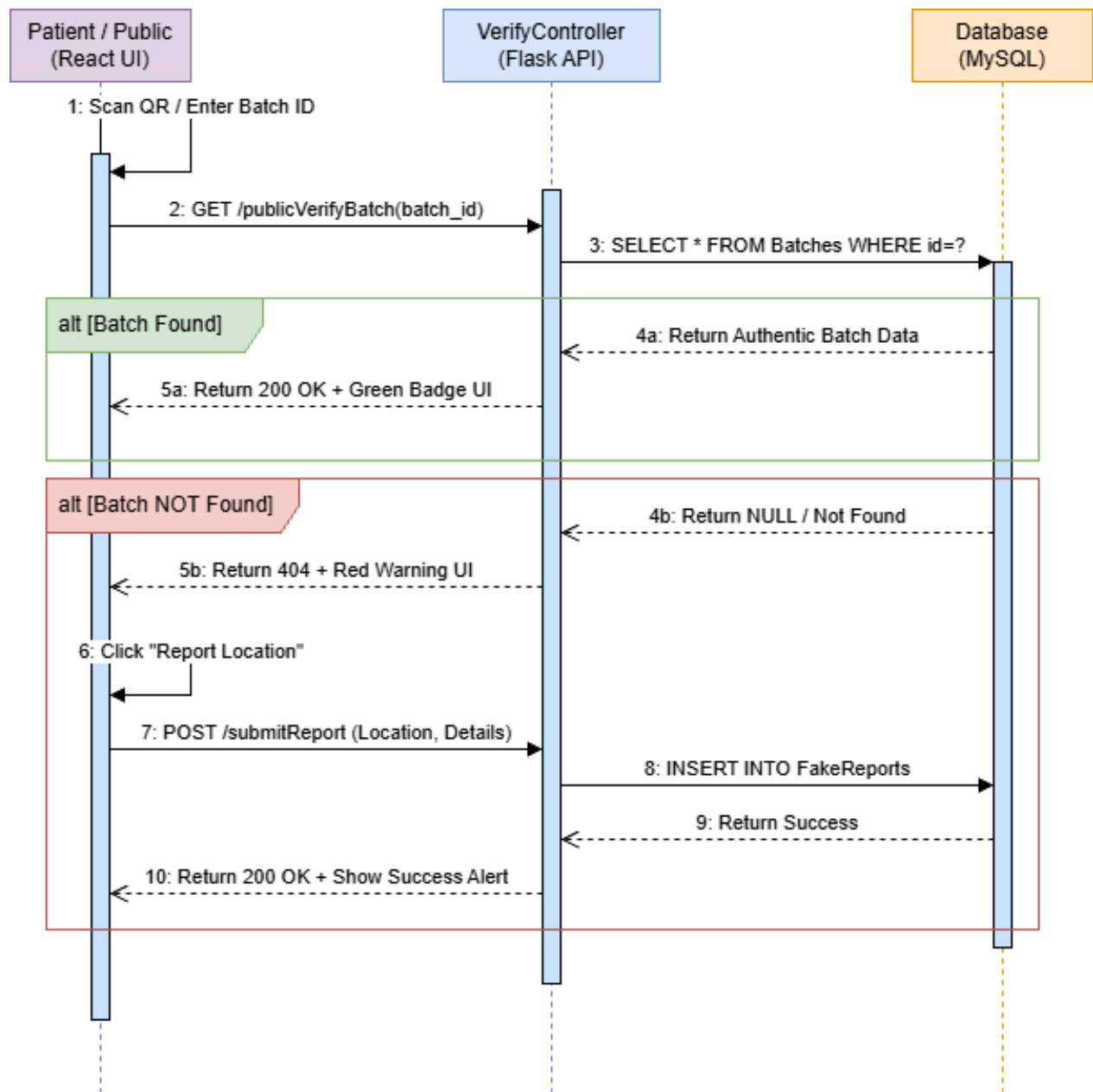


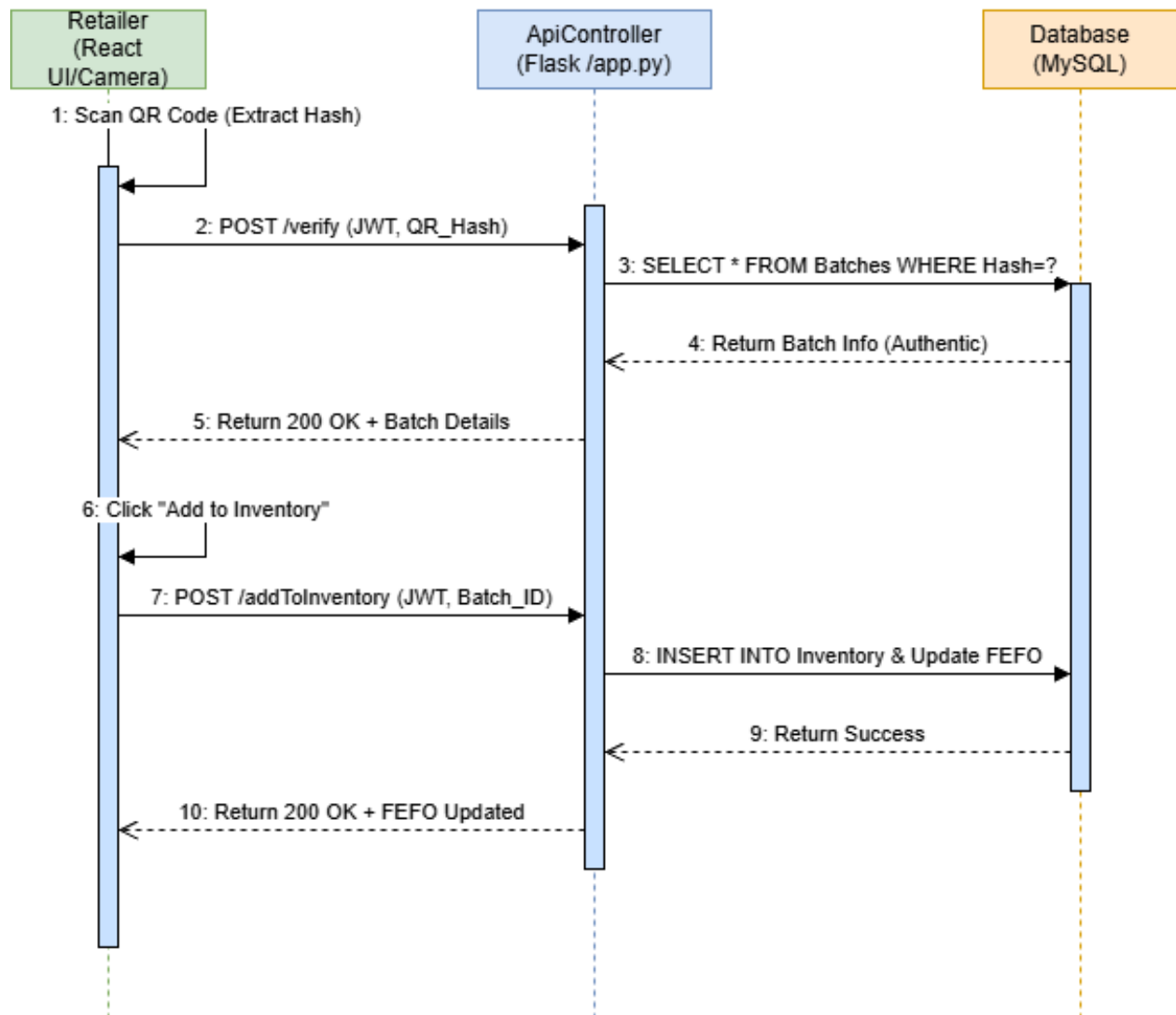
### 3.5 Sequence Diagrams

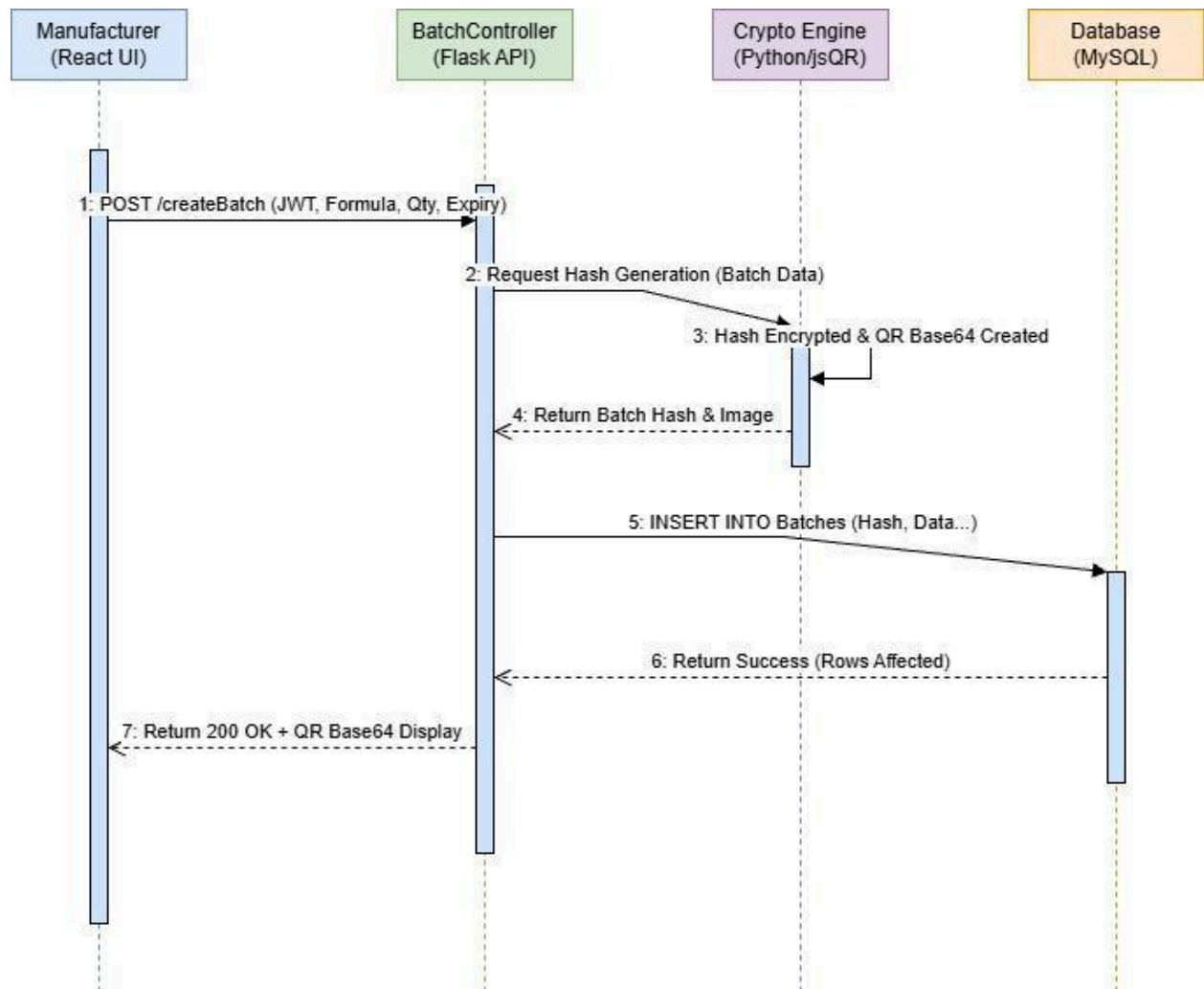




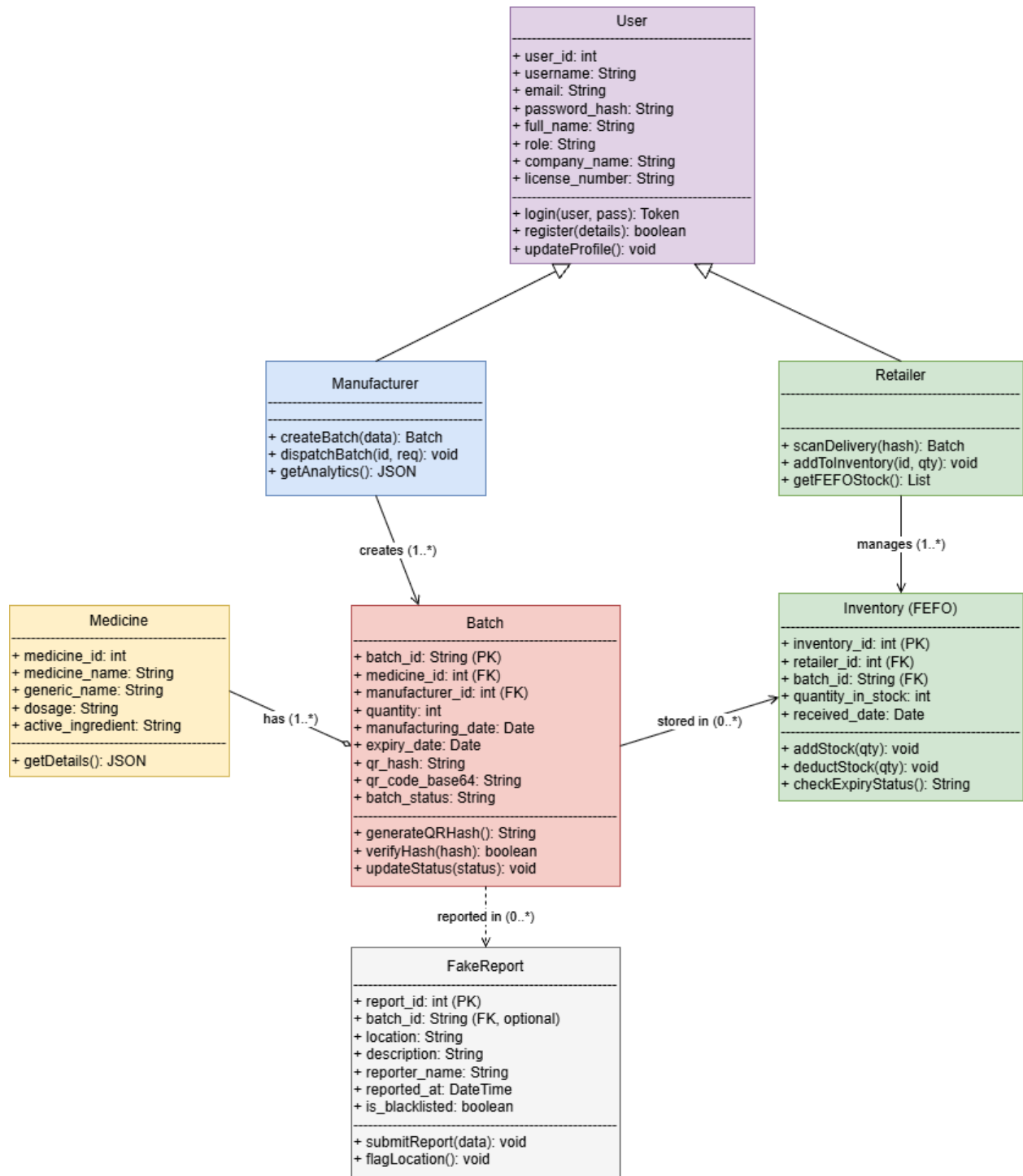




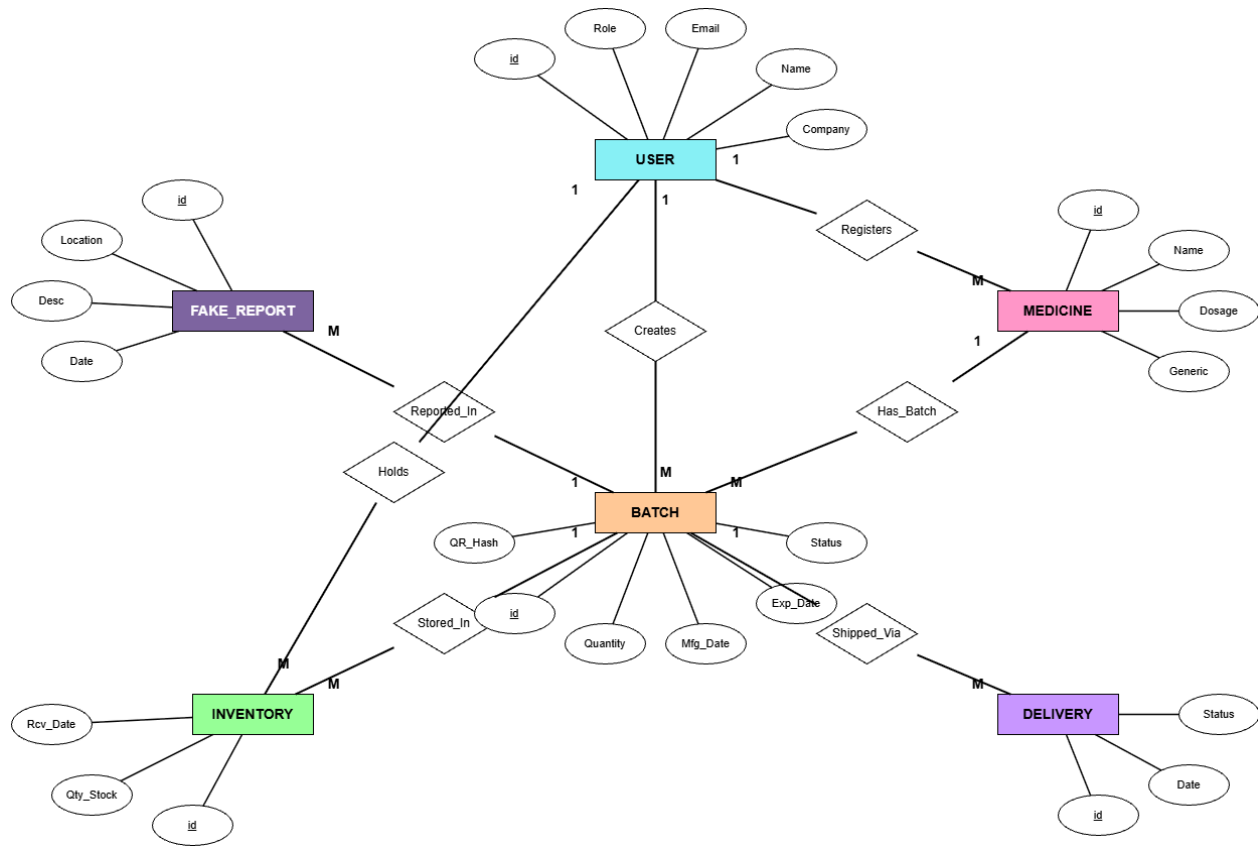




### 3.6 Class Diagram

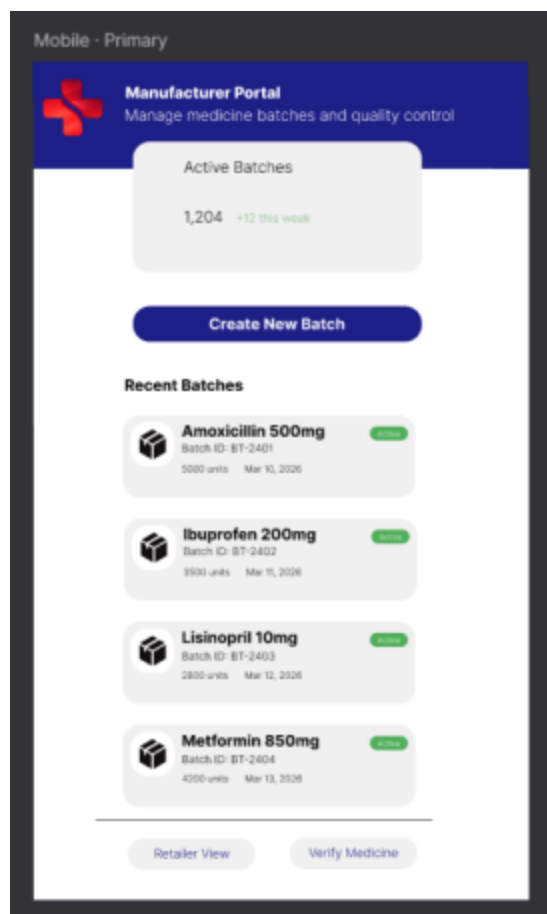


### 3.7 Database Design (ER Diagram / Schema)

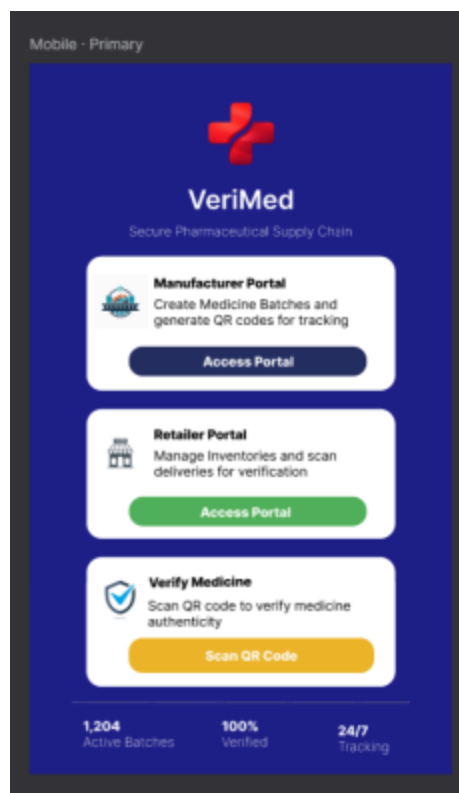


### 3.8 Interface Design (Wireframes / Prototypes)

<https://www.figma.com/site/uLuF2HHAdqqkOzU2yATWxM/Untitled?node-id=0-1&p=f&t=e9qxuGqRzLIDYzax-0>

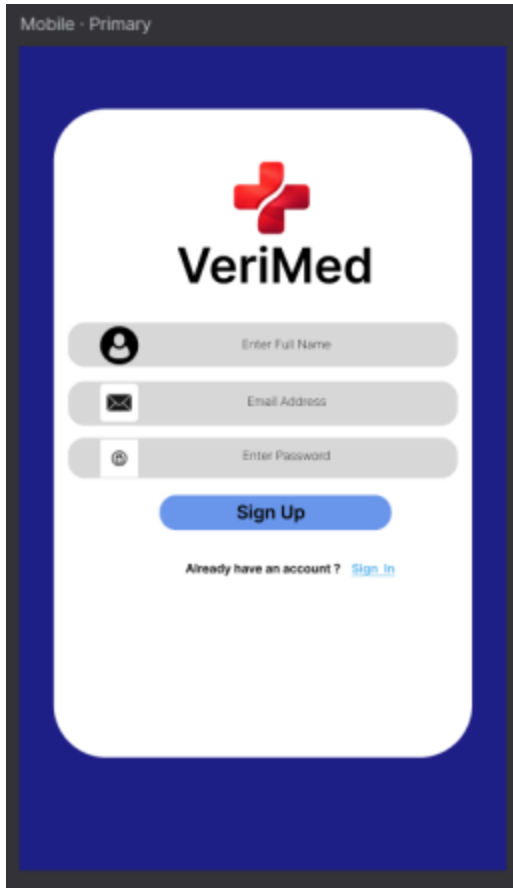


*Figure 3.8.1: Manufacturer Portal*



*Figure 3.8.2: Verimed Home Page*





*Figure 3.8.3: VeriMed SignUp page*

## Chapter 4: Implementation

### 4.1 Development Environment

This VeriMed system was developed in an organized design and development process which is designed to develop an efficient healthcare management application.

#### 4.1.1 Tools and Softwares

- **VS Code:** IDE which is used primarily used for coding
- **Figma:** Used for prototypes
- **Postman:** perform checks on RESTful APIs.
- **Git/GitHub:** It is used to maintain versions of code.

#### 4.1.2 Programming Languages and Components Development

- **HTML5 & CSS3:** Used to build website structure.
- **React.js:** It is used to develop a React-based front-end application with components and hooks like `useState` and `useEffect`.
- **Python (Flask):** It is used to develop the backend of an application.
- **MySQL:** MySQL database is used to store data properly in third normal form.

## **4.2 System Modules Description**

### **4.2.1 Manufacturer Module**

- The Manufacturer component aids in initiating the secure supply chain. This component comprises a secure login feature that provides access to a dashboard where the users can enter the information about the medicines. After submission, the system creates a hash function, which is then encoded to a Base64 QR code image and saved in the medicines batches table.

### **4.2.2 Retailer Module**

- The “Scan to Receive” process in the Retailer module makes use of jsQR to scan the barcode via the camera of the device and retrieve batch information. The collected information is later uploaded to the FEFO inventory dashboard, wherein the items nearing expiration are displayed in different colors.

### **4.2.3 Patient / Verification Module**

- This user-friendly module does not require any sign-in process for users to check the validity of medicines within seconds. It contains a scanner window along with an input section where users can enter their batch IDs. In case the batch ID is missing or out of date, then the module will notify the users about the same.

## **4.3 Frontend Implementation**

- The front-end was developed through React.js as a Single Page Application. This design of the website is supposed to adapt to various types of devices. The custom CSS modules were used to give the website its unique look. The website has also been made to navigate easily by implementing `react-router-dom`.

## **4.4 Backend Implementation**

- The backend is implemented using flask which is set with restful API, there are many significant functionalities some of them are
  - POST/Login for request authentication
  - POST/createBatch for unique batch generation
  - GET/publicVerifyBatch to verify the batch

## 4.5 Database Implementation

- Since we used mysql we were able to make the data base in 3nf form, the structure of our database includes a lot of tables like users, medicines etc, with proper foreign key maintained between the tables and other database rules are also followed.

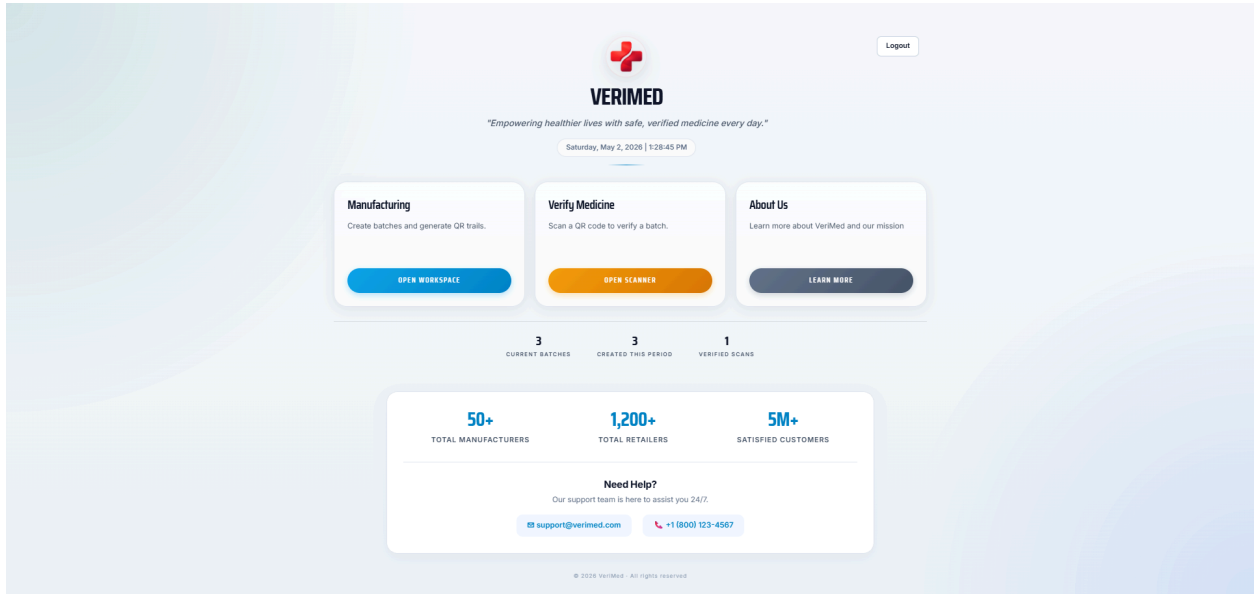
## 4.6 Integration of System Components

- The integration is carried out by making asynchronous API requests (via fetch or axios) from the React front-end to the Flask back-end. Upon form submission in HTML, the data is validated by the React front-end application, and then the data undergoes the process of being inserted into or updated in the MySQL database through the Flask script, with JSON feedback sent back to update the user interface accordingly.

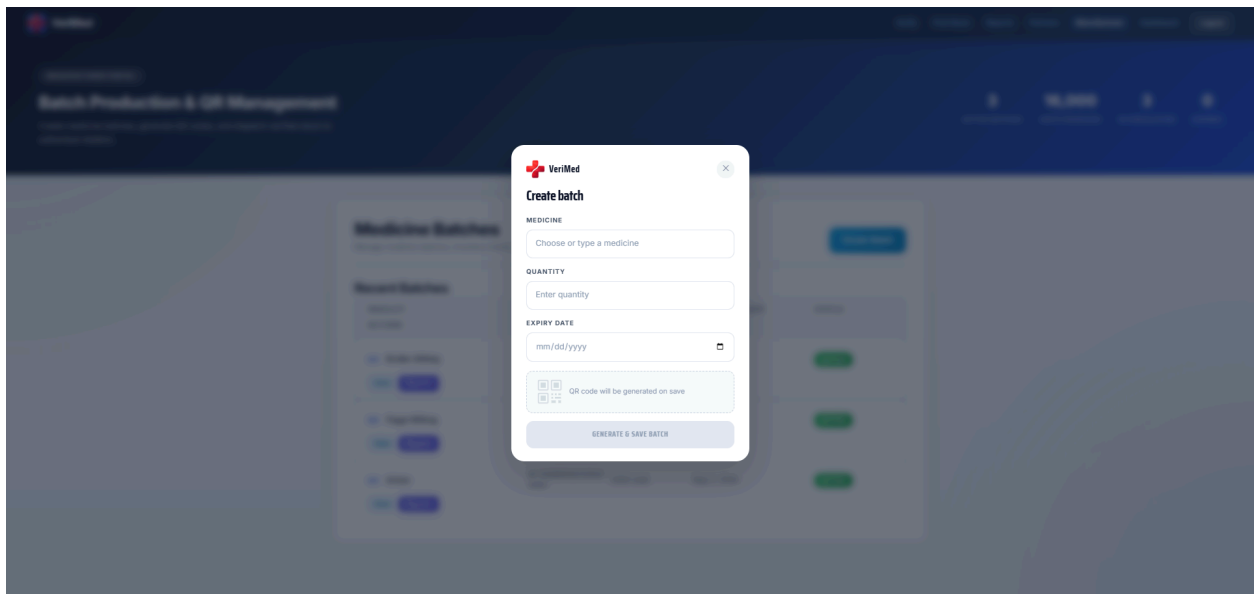
## 4.7 Key Functionalities

### 1. Features of Manufacturer Dashboard

- **Create Batch" Form:** This dashboard includes "Create Batch" form, through which manufacturer can enter data like Formula name, number of batches, expiration date, etc.
- **QR Code Generation:** Once this form is filled in, the back end generates unique code which becomes QR code on the dashboard itself.
- **List of Active Batches:** Other than this, there is a list of all the batches generated by manufacturer displayed on the dashboard.



**Figure 4.7.1: Manufacturer Homepage**



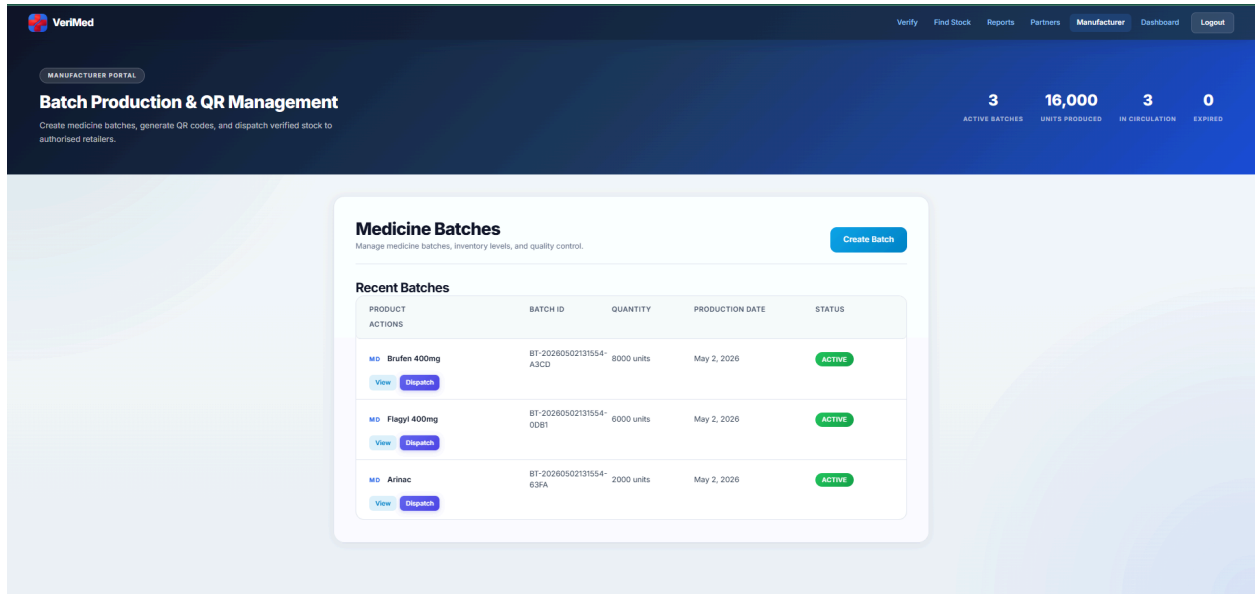
**Figure 4.7.2: Batch Creation**

The screenshot shows a web application interface with a dark blue header and sidebar. A modal window titled "Dispatch Batch" is open in the center. The modal has a red cross icon in the top left corner and a close button (X) in the top right. Below the title, there is a header bar with a small circular icon containing a red cross, followed by the text "BT-20260502131554-A3CD" and "Brufen 400mg". Below this, a note states: "Enter the retailer details where this batch is being sent. Customers will see this info when they search for this medicine." The form contains several input fields: "RETAILER / PHARMACY NAME" (with example text "e.g. MedPlus Pharmacy Lahore"), "CONTACT NUMBER" (with example text "e.g. +92-300-1234567"), "LOCATION / ADDRESS" (with example text "e.g. 24-B, Gulberg III, Lahore"), "WEBSITE / ORDER URL (optional)" (with example text "https://pharmacy.example.com"), "QUANTITY TO DISPATCH" (with example text "e.g. 500"), and "DELIVERY DATE" (with a date picker showing "mm/dd/yyyy"). At the bottom of the modal is a blue button labeled "CONFIRM DISPATCH".

**Figure 4.7.3:** Batch Dispatch

The screenshot shows the same web application interface. A modal window titled "Batch Details" is open in the center. The modal has a red cross icon in the top left corner and a close button (X) in the top right. Below the title, there is a section titled "MEDICINE INFORMATION" with the following details: "MEDICINE: Brufen 400mg", "BATCH ID: BT-20260502131554-A3CD", and "MANUFACTURER: Pharma Corp". Below this is a section titled "BATCH DETAILS" with the following details: "QUANTITY: 8000 units", "MANUFACTURING DATE: Sat, 02 May 2026 00:00:00 GMT", "EXPIRY DATE: Sun, 02 May 2027 00:00:00 GMT", and "STATUS: Active" (indicated by a green pill). At the bottom of the modal is a section titled "QR CODE" which contains a QR code.

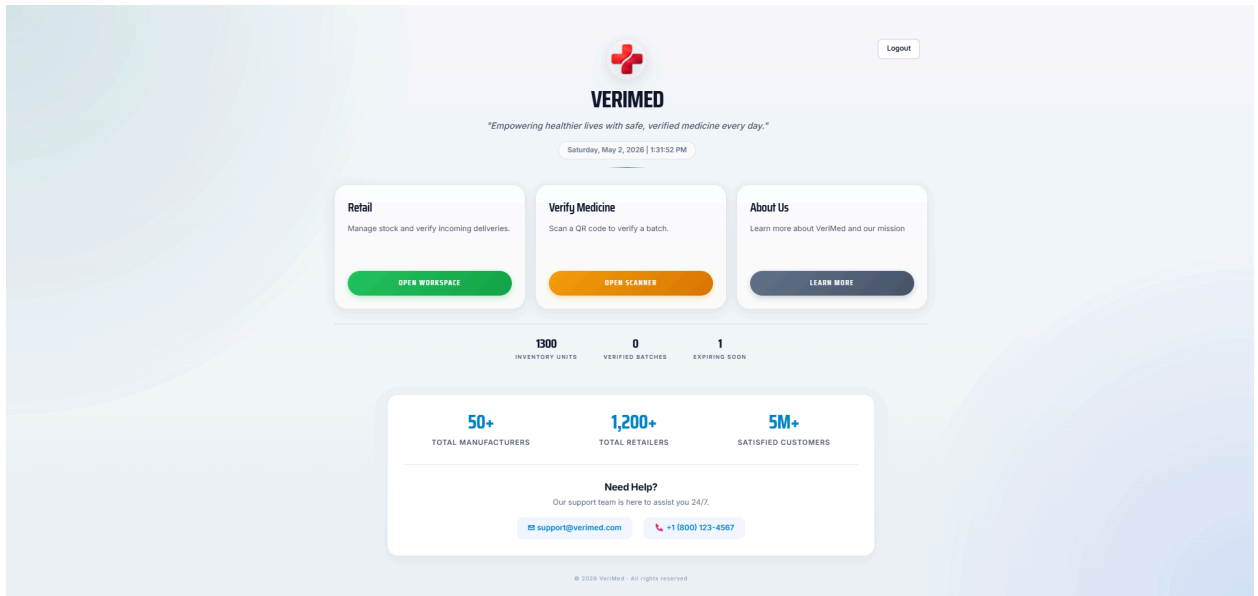
**Figure 4.7.4:** Batch Details



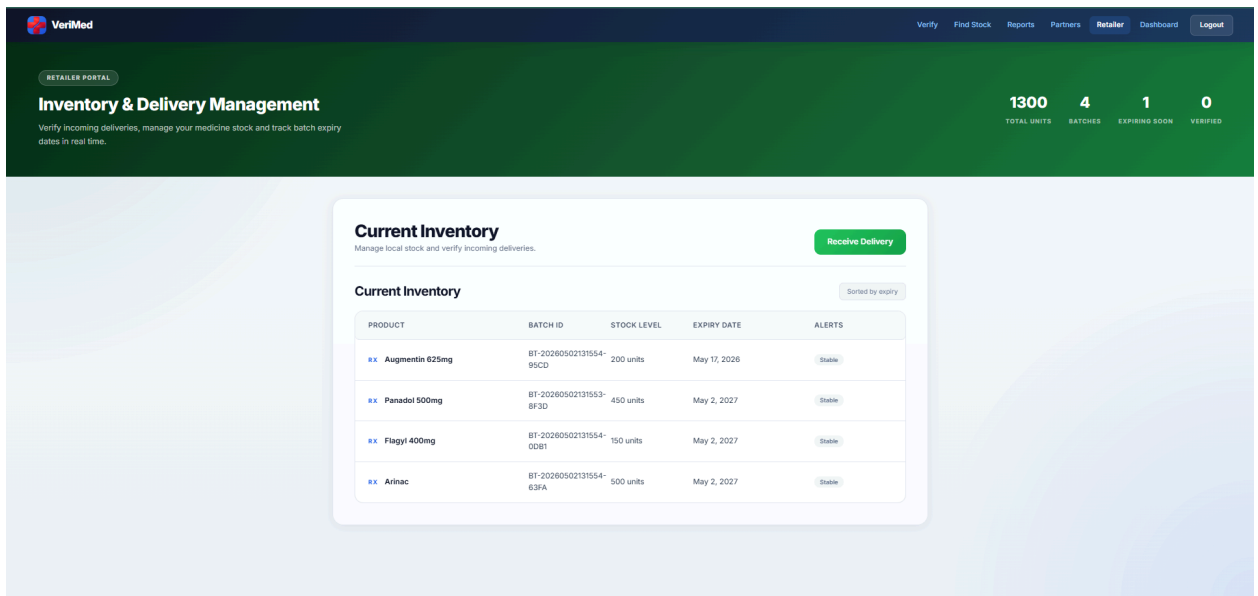
*Figure 4.7.5: Manufacturer Portal*

## 2. Retailer Dashboard & Its Features:

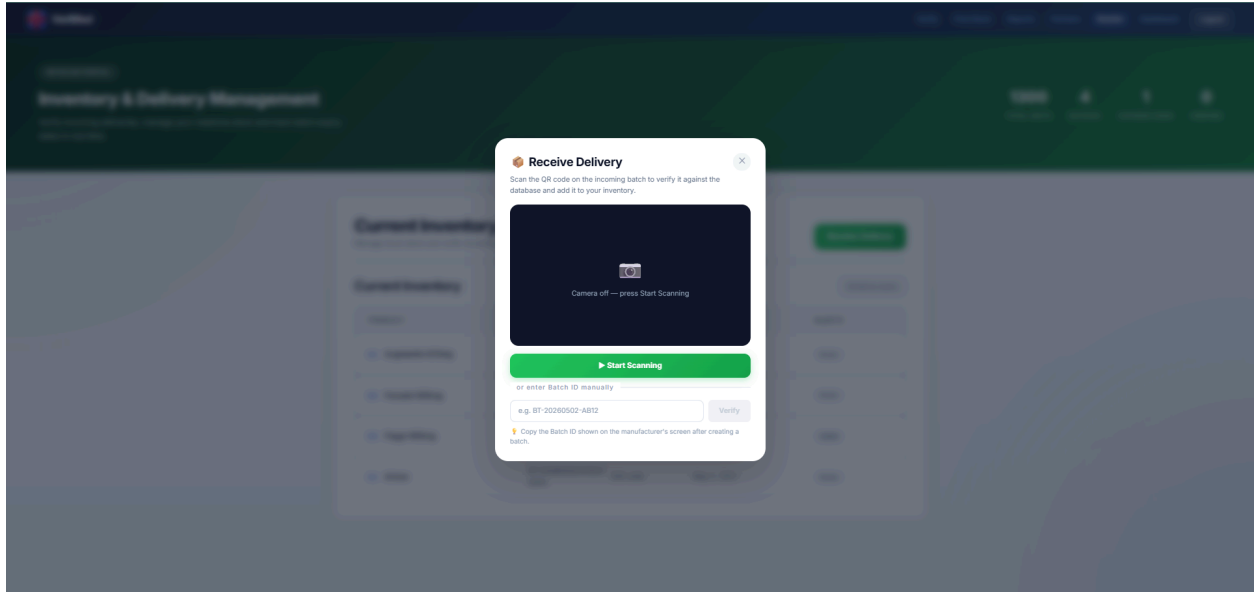
- Scan to Initiate Procedure:** Leveraging the functionalities offered by the jsQR plugin, it is necessary that the retailer scans the QR code created by the manufacturer utilizing the device camera. Immediately afterward, a fast “handshake” session starts whereupon authentication of the product occurs, following which it gets added to the list of inventories.
- FEFO Based Inventory Control System:** The products are classified in accordance with the FEFO rules, and the medicines in the list show the following status depending on their classification:
  - Green:** A valid item with sufficient shelf life.
  - Yellow:** Expiry Warning within six months.
  - Red:** Expired stock that cannot be sold as per system requirements.
- Stock Analytics:** Retailers get access to the total number of units sold compared to units in stock.



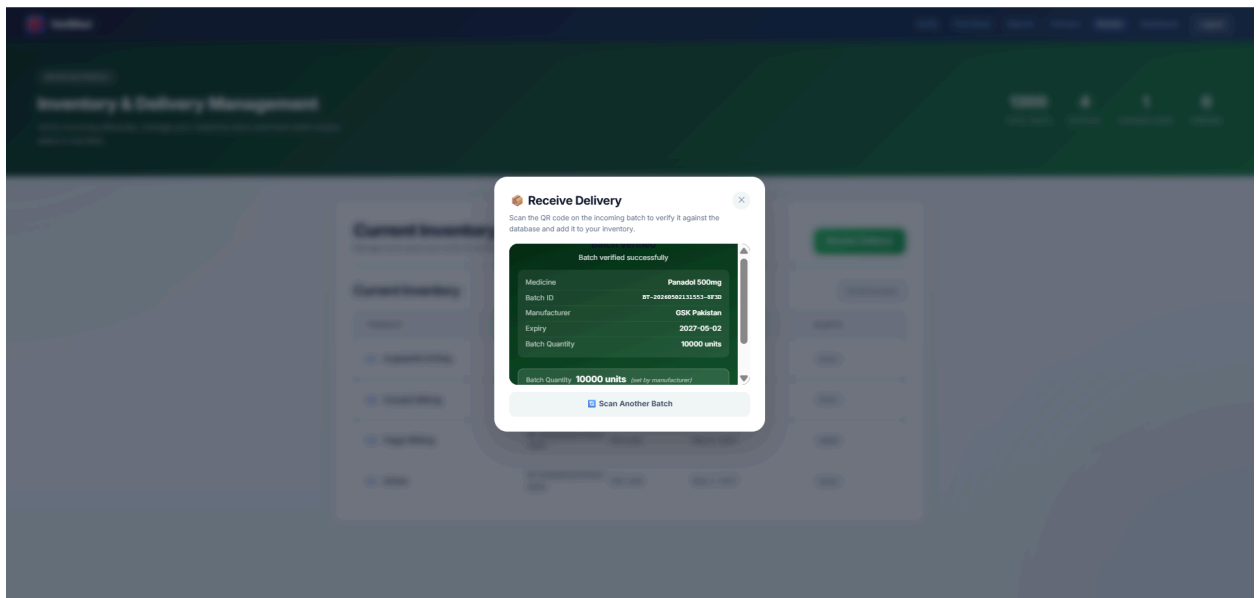
**Figure 4.7.6:** Retailer's Homepage



**Figure 4.7.7:** Retailer Portal



*Figure 4.7.8: Receive Delivery*



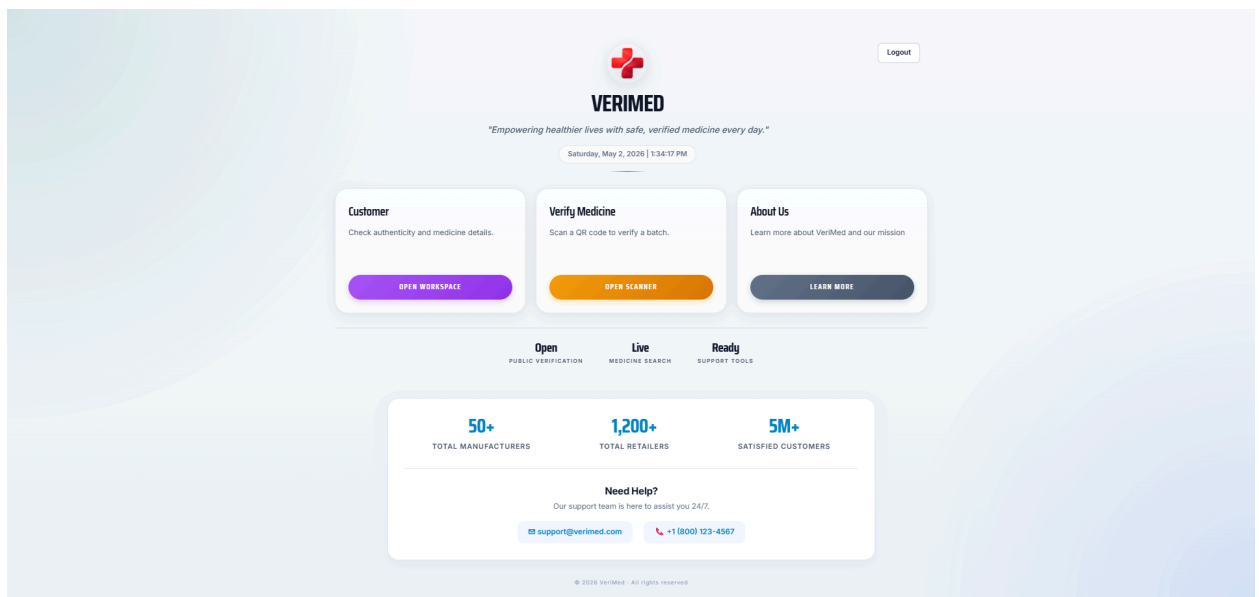
*Figure 4.7.9: Verification Successful*

### 3. Customer Dashboard and Functions:

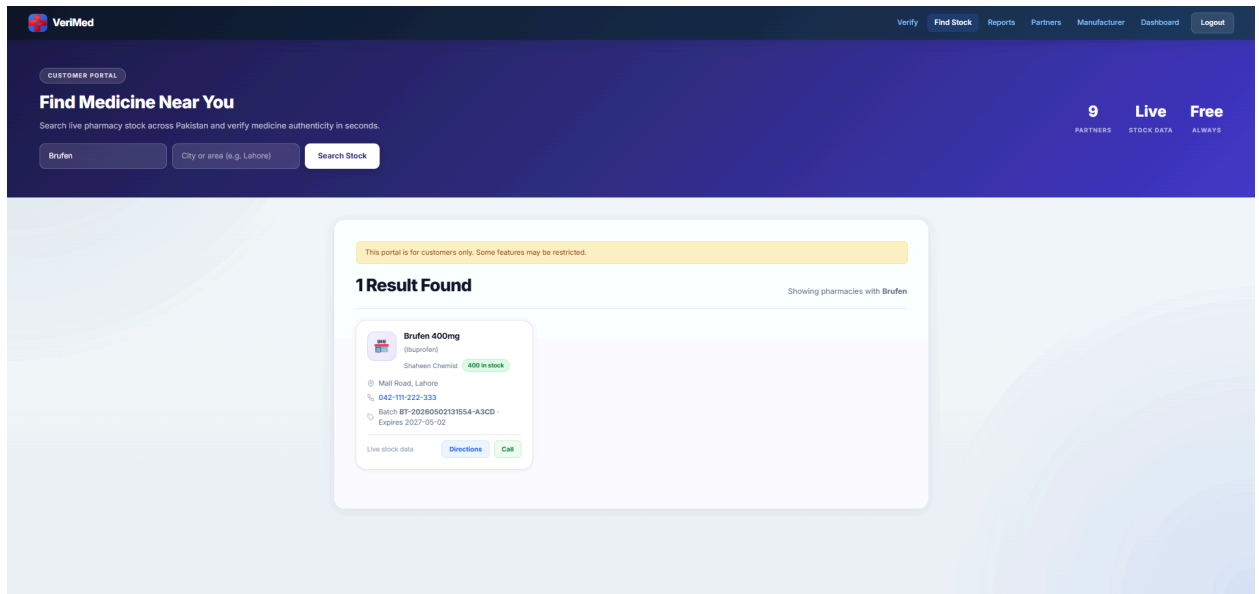
- Verify Medicine Public Interface:** This screen will present customers with a user-friendly Verify Medicine screen that consists of an in-built camera scanner as well as a Batch ID entry box.



- **Provenance Details:** Successful scan results in displaying a "Genuine" badge along with full provenance details including the name of the manufacturer, license details and expiration date of the medicine.
- **Warning Screen:** In case there is any anomaly in the code inputted in the system, the user interface automatically switches to a warning screen that contains a reporting form whereby the consumer can report details of the pharmacy location and store name which will be captured in the qr\_verifications table.
- **Medicine Finder Screen:** Customers are able to find out which pharmacies have specific medicines in stock near them.

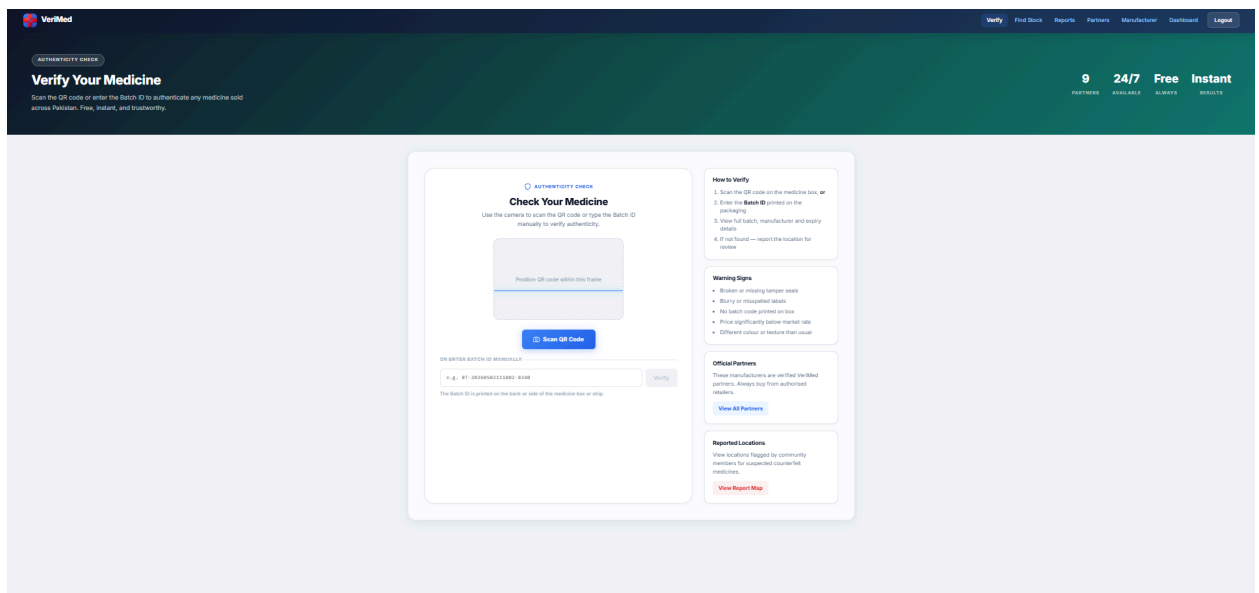


**Figure 4.7.10:** Customer's Homepage

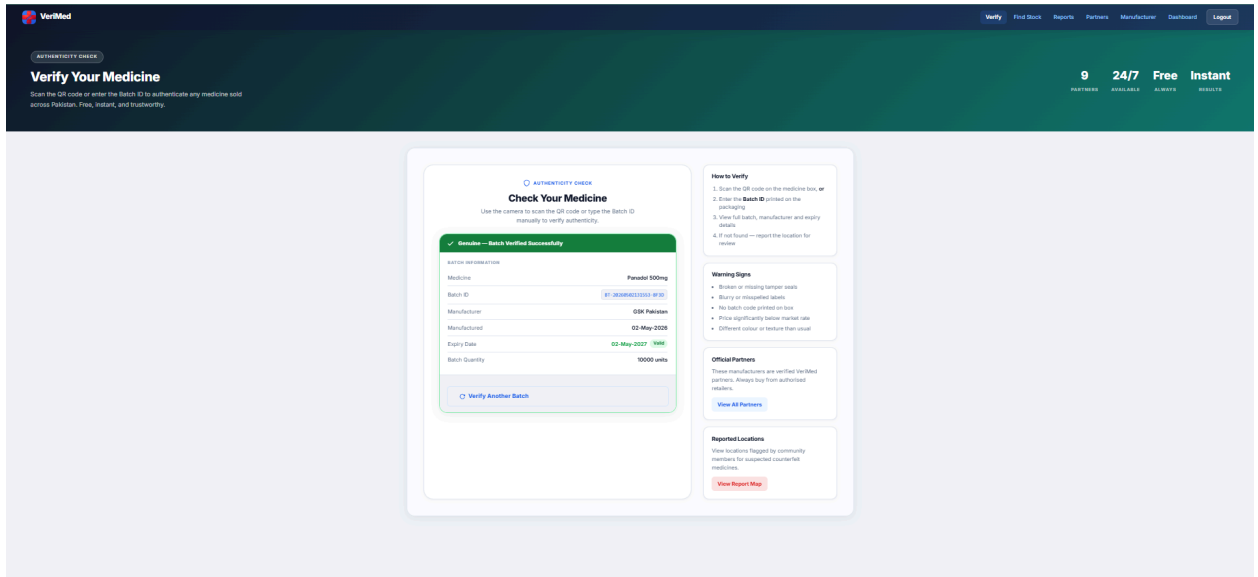


*Figure 4.7.11: Search Medicine*

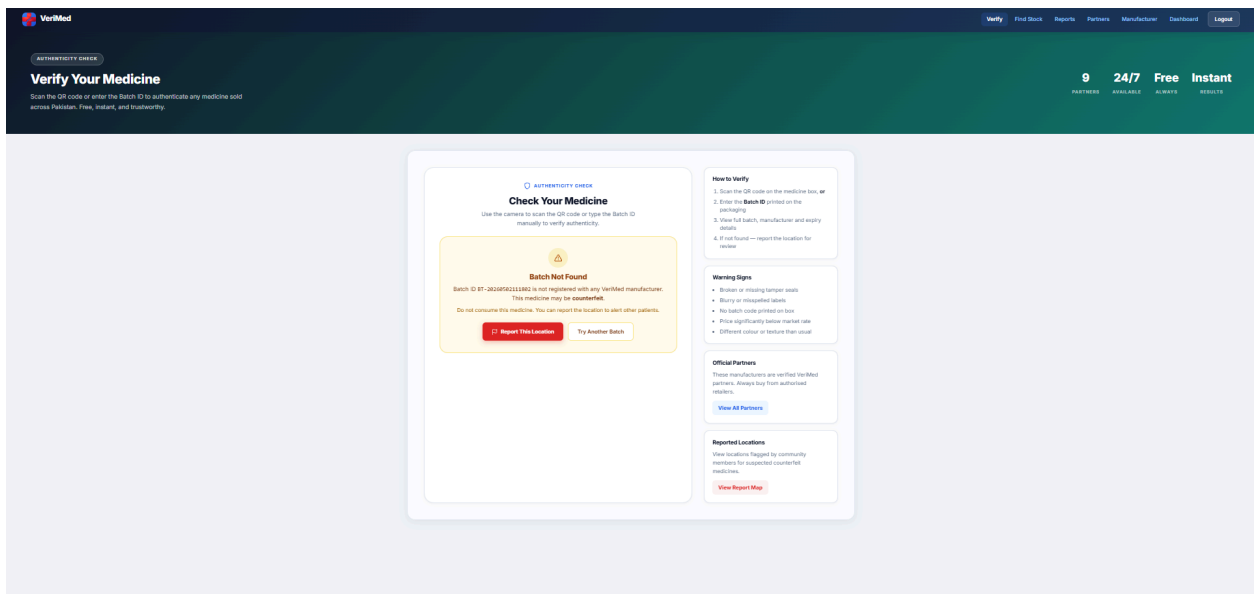
4. **Public Verification Screen:** An on-the-go verification screen that instantly lets the consumer know whether the item is a “Genuine” or a “Counterfeit.”



*Figure 4.7.12: Public verification portal*



**Figure 4.7.13: Successful Verification**



**Figure 4.7.14: Unsuccessful Verification**

5. **Crowdsourced Reporting:** A reporting feature in the module that allows people to flag suspicious areas where purchases are made.

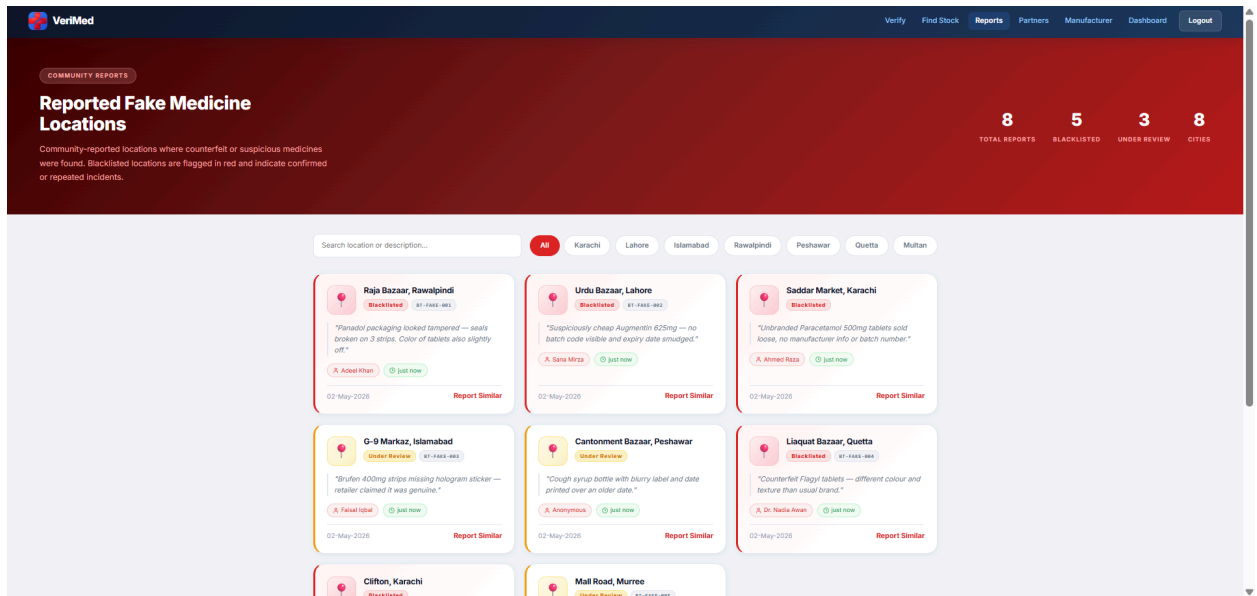


Figure 4.7.15: Reported Fake Medicine Locations

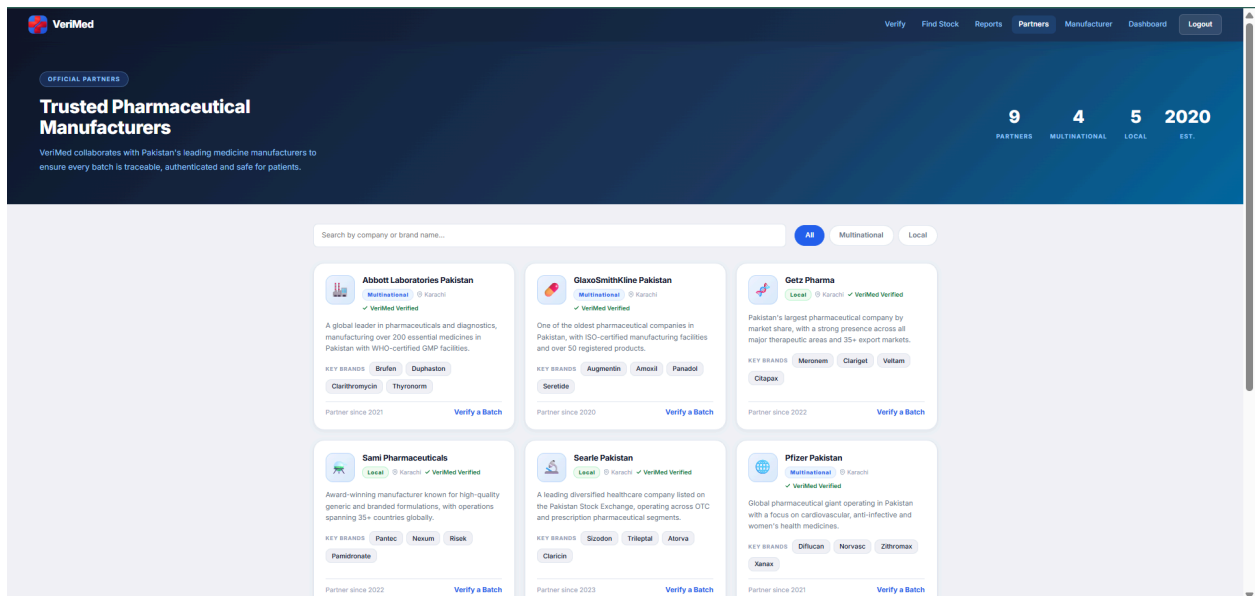


Figure 4.7.16: Trusted Pharmaceutical Partners

## Chapter 5: Testing & Evaluation

### 5.1 Testing Strategy

The testing strategy for VeriMed was planned such that the integrity of the medical supply chain is validated by ensuring that all the components of the app function well and provide accurate results.

### 5.1.1 Testing Levels

- **Unit Test:** The tests focused on checking the functionality of the functions in the Flask back-end such as the hashing of the message and the extractBatchId tool.
- **Integration Test:** This test ensured the proper functioning of the connection between React front-end and MySQL database using Flask API services, whereby the form data input into forms (Signup.jsx, Manufacturer.jsx) will be saved in the database.
- **System Testing:** The system test was done using the full flow of batch creation by the manufacturers to final authentication process of the batches by patients.

### 5.1.2 Methodology

- **Functional Testing:** The testing procedure that ensured whether functionalities such as QR scanner, role-based login, etc., met the requirements.
- **Boundary Value Analysis (BVA):** The method that was adopted to test the critical parameters like Batch Size and Expiry Date.

## 5.2 Test Cases and Results

The following test cases were executed to validate the system's core functionalities.

| Test ID | Test Description                  | Expected Result                                | Result |
|---------|-----------------------------------|------------------------------------------------|--------|
| TC-01   | User Login with valid credentials | Successful redirect to role-specific dashboard | Pass   |
| TC-02   | Batch Creation with valid data    | QR code generated and saved to DB              | Pass   |
| TC-03   | QR Code Scan (Authentic Batch)    | System displays "Genuine" status and details   | Pass   |
| TC-04   | QR Code Scan (Fake/Invalid Hash)  | System displays "Counterfeit" warning          | Pass   |
| TC-05   | Inventory FEFO Sorting            | Stock sorted by earliest expiry date           | Pass   |

|       |                             |                                                  |      |
|-------|-----------------------------|--------------------------------------------------|------|
| TC-06 | Search Medicine by City     | Returns list of local verified retailers         | Pass |
| TC-07 | Submit Fake Report          | Report details logged in qr_verifications        | Pass |
| TC-08 | Unauthorized Access to /mfg | Redirect to login / 403 Forbidden                | Pass |
| TC-09 | Password Reset Request      | Confirmation message "Reset Link Sent" displayed | Pass |
| TC-10 | Empty Form Submission       | Validation error message displayed               | Pass |

### Boundary Value Analysis (BVA) for Critical Inputs

| Field       | Input Type    | Value Tested | Expected Result                 | Status |
|-------------|---------------|--------------|---------------------------------|--------|
| Quantity    | Minimum       | 1            | Accepted                        | Pass   |
| Quantity    | Zero/Negative | 0, -5        | Error: "Invalid Quantity"       | Pass   |
| Expiry Date | Past Date     | 2023-01-01   | Error: "Date cannot be in past" | Pass   |
| Expiry Date | Far Future    | 2050-12-31   | Accepted                        | Pass   |

## 5.3 Validation of System

According to the results of testing, the system is working properly in respect of the following functionalities:

- **FR-01 (Authentication):** The users are able to authenticate and get access to their roles using role based access along with secure session management technique (JWT).
- **FR-02 (Traceability):** Hash values are generated for each batch which are traceable and successfully validated.
- **FR-03 (Inventory Management):** Validation through the automated FEFO badge system and inventory stock updates.
- **FR-04 (Public Safety):** Validation through the counterfeit warning and location reporting system.

## 5.4 Performance Testing

For the performance tests, the following functions of the system were assessed for their response time:

- **Database Queries:** For searching the pharmacy locally and for inventory searches which were all found to be less than 0.5 seconds thanks to optimized MySQL indexing.
- **QR Code Creation:** Conversion of batch details into base64 image in less than one second.

## 5.5 Feedback from Users

Based on the initial feedback received internally, the “Glassmorphism” user interface design enhanced the usability for the retailers who had large inventories. External feedback from users will be sought in the upcoming development stages to enhance the community reporting feature.

# Chapter 6: Conclusion & Future Work

## 6.1 Summary of the Project

The problem of fake drugs, which is one of the most important problems on a global scale, is solved through the VeriMed initiative using an information system based on cryptography of QR codes together with a central relational ledger. The solution lies in connecting the missing link between producers, retail stores, and end users through a reliable database of unique digital IDs of every drug unit, allowing for instant identification through a smartphone camera.

## 6.2 Achievements

The project achieved many of the major components and capabilities outlined in this report:

- **Successful Portal Deployment:** There are three completely separate and entirely discrete role based portals (Manufacturer, Retailer, and Customer), each of them are with business logic and security specific to that component identified.

- **Secure Cryptographic Ledger:** We have implemented a very strong backend that generates batch hashes, and it completely automatically saves it to a 3NF MySQL database so nothing can be lost.
- **Real-time Verification:** Implemented jsQR for client-side scanning, enabling a frank check instantaneously on the order of decimals without requiring proprietary hardware.
- **Automated FEFO Inventory:** An automated "First-Expired-First-Out" management system has been implemented and it is completely integrated into the pharmacy, which alerts pharmacists to upcoming expirations, directly impacting patient safety.

## 6.3 Limitations

While the current iteration of VeriMed meets all core functional requirements, certain limitations exist:

- **Localhost Deployment:** Current deployment is limited to a local network environment, thereby making it not immediately usable in any global supply chain.
- **Centralized Database:** Entire architecture is completely built on a digitally centralized MySQL database, which can lead to a single point and very serious threat of administrative-level data tampering.
- **Manual Reporting:** The reporting format works, but instead of being automated by geolocation it depends on customers.

## 6.4 Future Enhancements

The first exercise aimed at developing new features to make VeriMed a promising, more resilient and scalable platform:

- **Cloud Deployment:** The deployment of the Flask backend, and the MySQL database in cloud service providers like AWS or Heroku to provide extensive global accessibility and high end availability.
- **Decentralized Blockchain Ledger:** Replace with Blockchain-based, decentralized distributed ledger from batch history to eliminate single points of failure and provide an immutable, tamper-proof history of every medicine batch
- **Machine Learning Integration:** Predictive analytics to identify possible counterfeit "hotspot" based on crowdsourced reporting data.
- **Mobile App Expansion:** Create native Android and iOS apps to deliver a seamless, optimized scanning experience to consumers in rural markets.

## Appendix A: Source Code Snippets

Below we've attached all the snippets that represent the core logical modules of VeriMed system, these are extracted from the already implemented Flask backend and React frontend.



## A.1 Backend: Cryptographic Batch & QR Generation (app.py)

r physical labeling.

Python

```
@app.route('/createBatch', methods=['POST'])
@jwt_required()
def create_batch():
    data = request.get_json()
    batch_id = data.get('batch_id')
    medicine_id = data.get('medicine_id')
    manufacturer_id = get_jwt_identity() # Secured via JWT
    expiry_date = data.get('expiry_date')

    # Generate cryptographic hash for the QR code
    qr_data = f"Batch: {batch_id}|Med: {medicine_id}|Exp: {expiry_date}"
    qr_hash = hashlib.sha256(qr_data.encode()).hexdigest()

    # Generate QR Code Image as Base64
    qr = qrcode.make(qr_hash)
    buffered = BytesIO()
    qr.save(buffered, format="PNG")
    qr_base64 = base64.b64encode(buffered.getvalue()).decode()

    # Save to medicine_batches table
    new_batch = MedicineBatch(
        batch_id=batch_id,
        medicine_id=medicine_id,
        manufacturer_id=manufacturer_id,
        qr_code_base64=qr_base64,
        batch_status='active'
    )
    db.session.add(new_batch)
    db.session.commit()
    return jsonify({"message": "Batch Created", "qr": qr_base64}), 201
```

## A.2 Database: FEFO Inventory Sorting Query (database.sql)

```
-- Query to fetch inventory sorted by expiration date (FEFO)
SELECT
    m.medicine_name,
    b.batch_id,
    ri.quantity_in_stock,
    b.expiry_date,
    DATEDIFF(b.expiry_date, CURDATE()) AS days_to_expiry
FROM retailer_inventory ri
JOIN medicine_batches b ON ri.batch_id = b.id
JOIN medicines m ON b.medicine_id = m.id
WHERE ri.retailer_id = %s AND ri.quantity_in_stock > 0
ORDER BY b.expiry_date ASC;
```

### A.3 Frontend: Real-Time QR Scanner Integration (Retailer.jsx)

```
const scanQRCode = () => {
    if (videoRef.current.readyState === videoRef.current.HAVE_ENOUGH_DATA) {
        const canvas = canvasRef.current;
        const context = canvas.getContext("2d");
        context.drawImage(videoRef.current, 0, 0, canvas.width, canvas.height);

        const imageData = context.getImageData(0, 0, canvas.width, canvas.height);
        const code = jsQR(imageData.data, imageData.width, imageData.height);

        if (code) {
            console.log("Found QR code:", code.data);
            handleVerification(code.data); // Sends hash to Flask API
        }
    }
    requestAnimationFrame(scanQRCode);
};
```

## Appendix B: User Manual

## **B.1 For Manufacturers: Creating a Batch**

Login:

You need to login through the Manufacturers portal via your credentials

Data Input:

Go to the Create Batch page, enter the formula, batch size, and expiration date into it.

Create QR:

Click on the “Generate QR” button and download the image file. Print the QR code on the package of your batch.

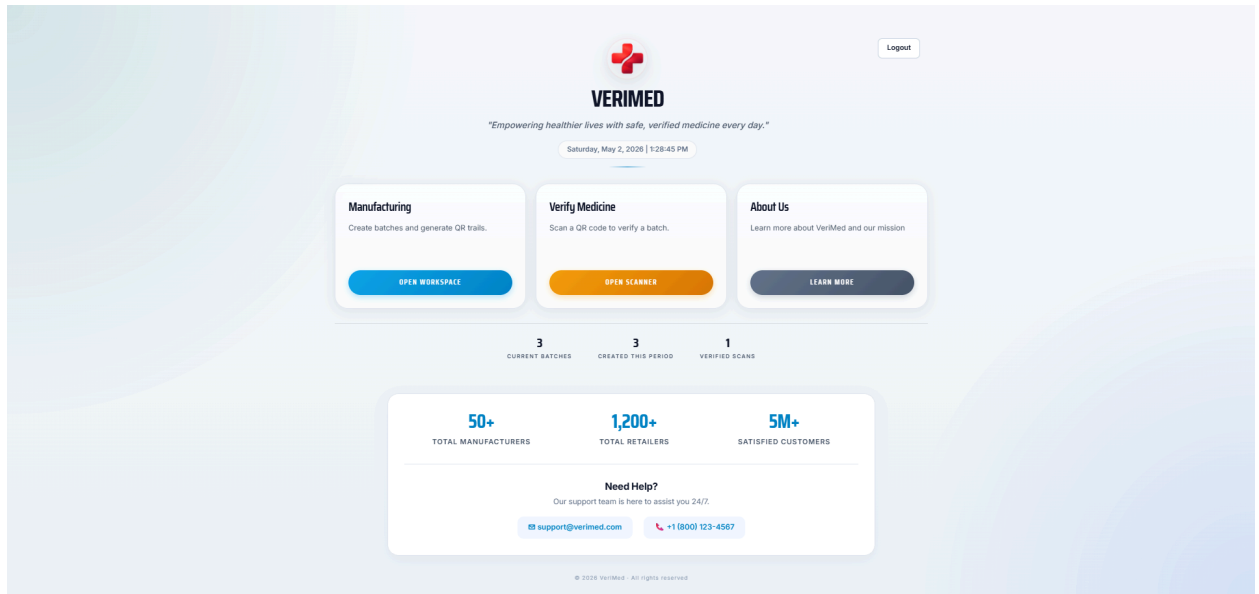
## **Section B.2 For Retailers: Accepting Stock**

- **Scanning QR:**  
Select the Retailer Dashboard and click the Accept Delivery button.
- **Verification:**  
By using the QR Code. The scanner will emit a green light if the batch is registered.
- **Inventory Update:**  
Click on “Add to Inventory” to update the FEFO stock inventory automatically.

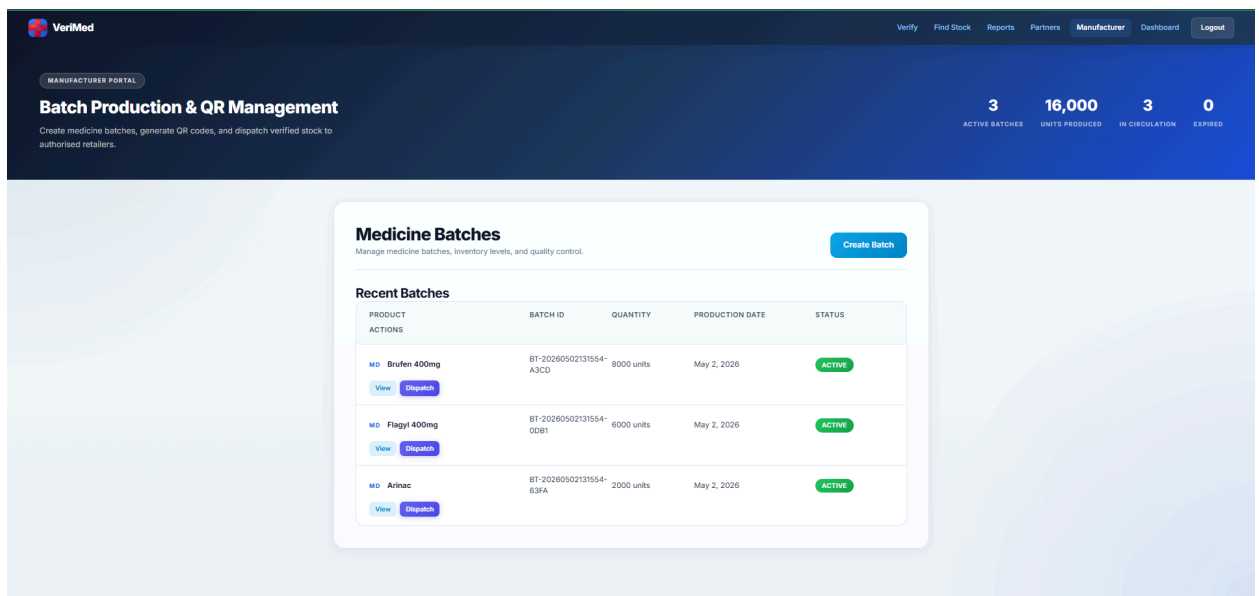
## **Section B.3 For Patients: Validating Medication**

- **Scanner**  
Visit the publicly accessible VeriMed website on your phone’s web browser without logging in.
- **Scan Medicine**  
Scan the barcode of the medicine.
- **Verification**  
The verification of the medicine will be done by analyzing its manufacture and expiry date.

## **Appendix C: Screens / UI Designs**



**Figure C-1: VeriMed Home Page for Manufacturer**



**Figure C-2: Manufacturer Portal**

**VeriMed**

**Create batch**

**MEDICINE**

Choose or type a medicine

**QUANTITY**

Enter quantity

**EXPIRY DATE**

mm/dd/yyyy

QR code will be generated on save

GENERATE & SAVE BATCH

**Figure C-3:** Medicine batch creation for Manufacturer

**VeriMed**

**Dispatch Batch**

BT-29240502131554-3CB Brufen 400mg

Enter the retailer details where this batch is being sent. Customers will see this info when they search for this medicine.

**RETAILER / PHARMACY NAME**

e.g. MedPlus Pharmacy Lahore

**CONTACT NUMBER**

e.g. +92-300-1234567

**LOCATION / ADDRESS**

e.g. 24-B, Gulberg III, Lahore

**WEBSITE / ORDER URL (optional)**

https://pharmacy.example.com

**QUANTITY TO DISPATCH**

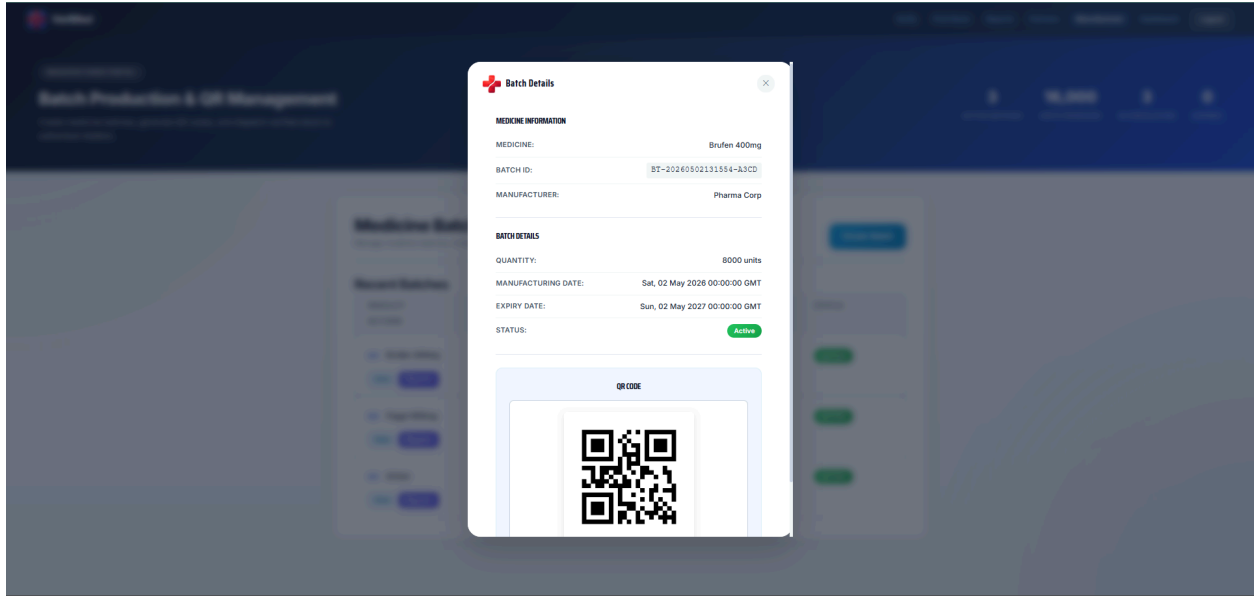
e.g. 500

**DELIVERY DATE**

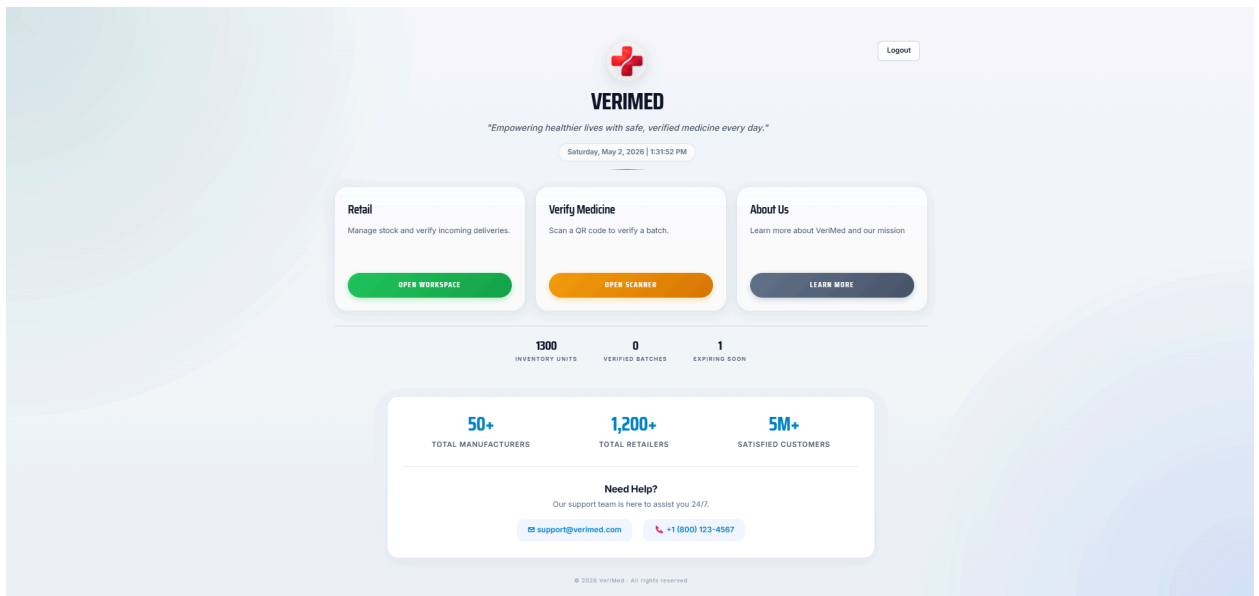
mm/dd/yyyy

CONFIRM DISPATCH

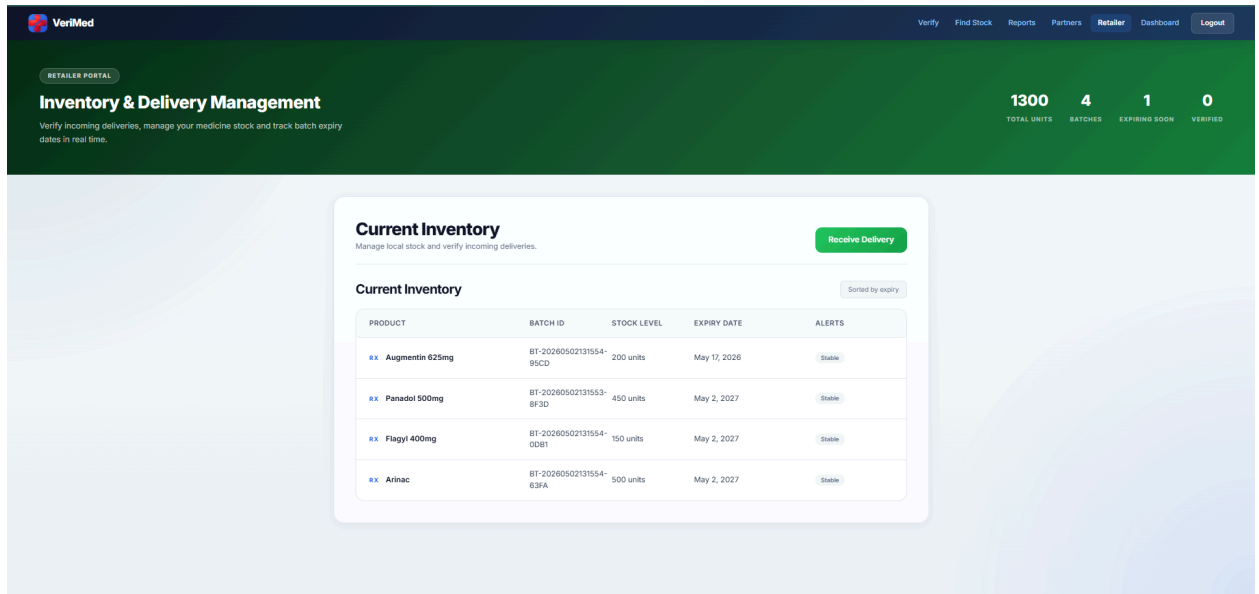
**Figure C-4:** Batch Dispatch for Manufacturer



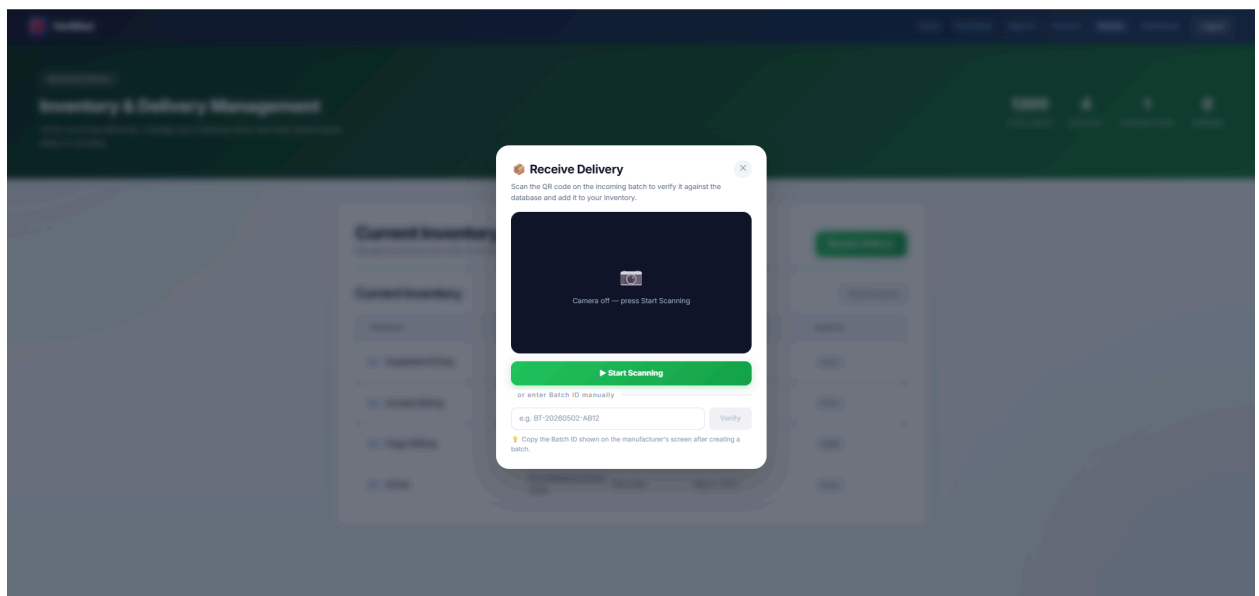
**Figure C-5: Batch Details Display for Manufacturer**



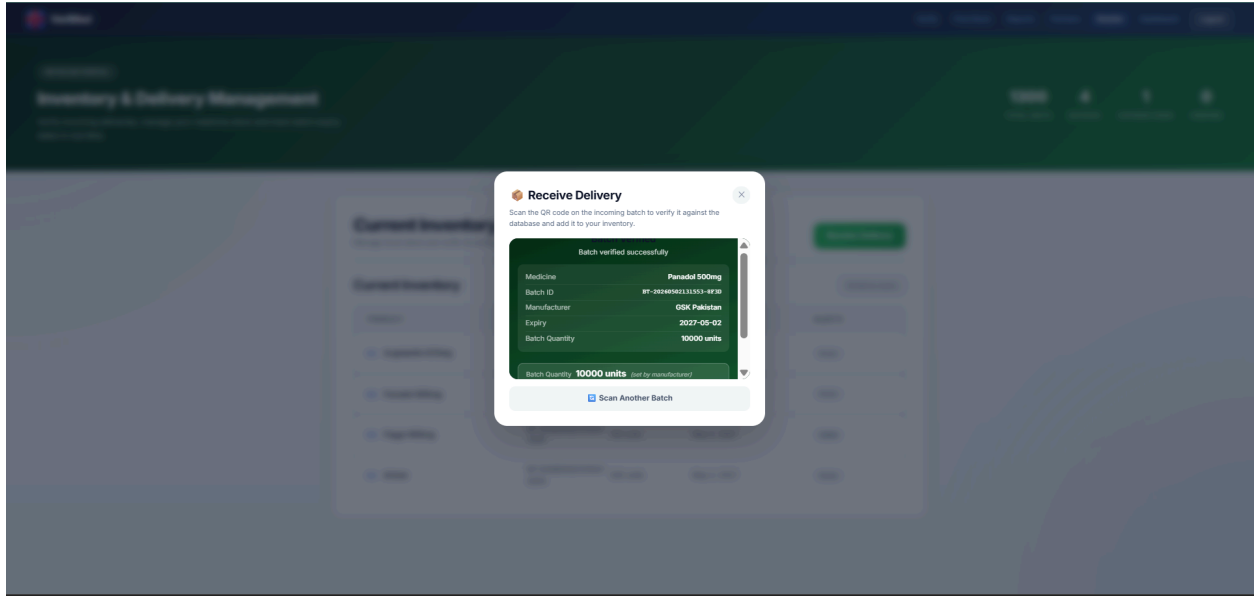
**Figure C-6: VeriMed Homepage for Retailer**



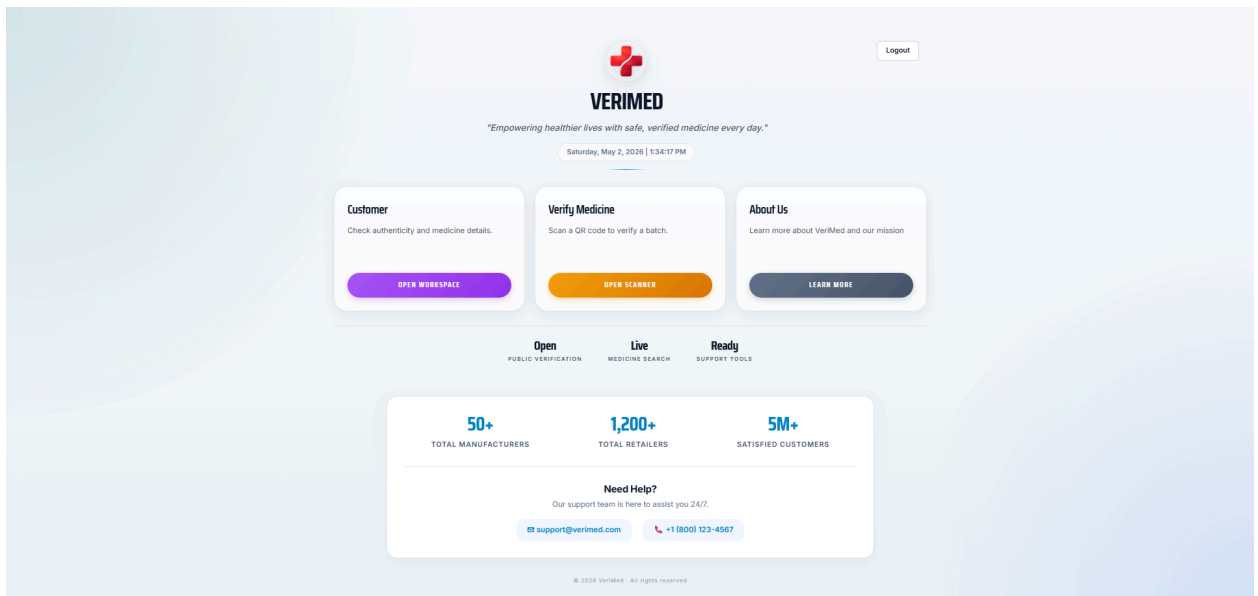
**Figure C-7: Retailer Portal**



**Figure C-8: Retailer Delivery Verification**

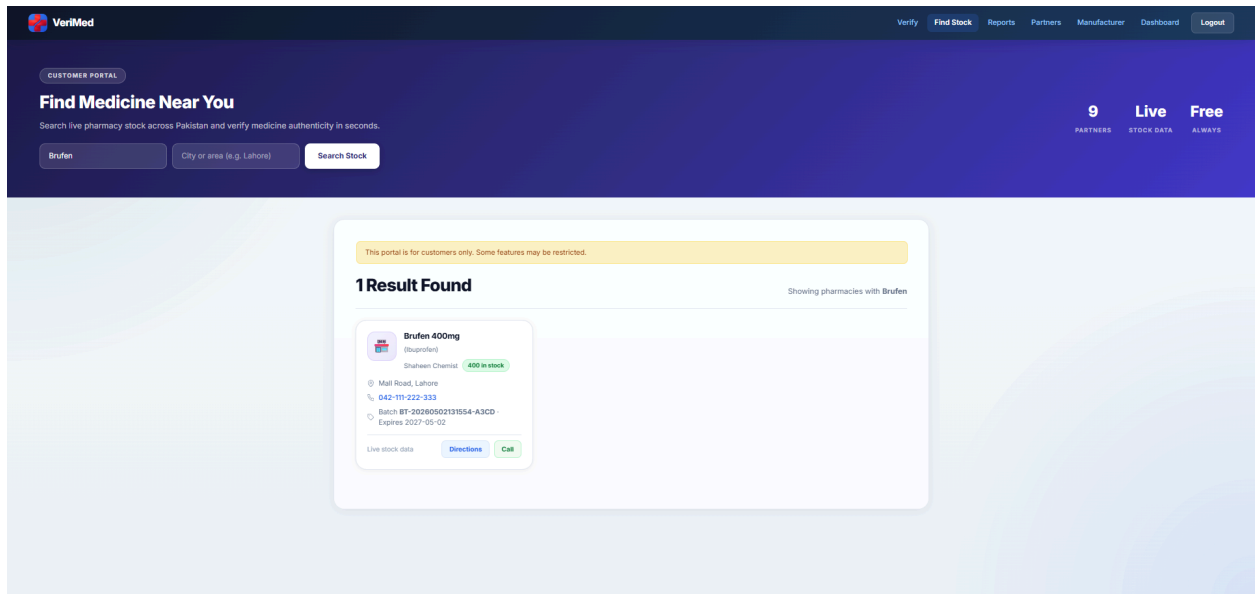


**Figure C-9:** Retailer Delivery Verification

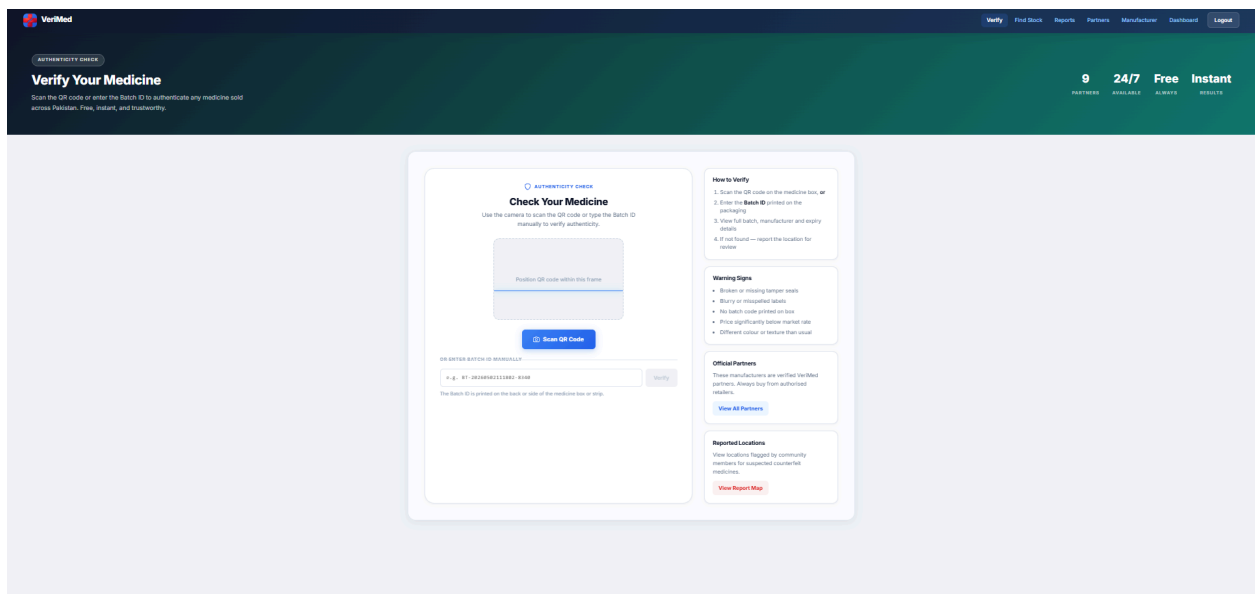


**Figure C-10:** VeriMed Homepage for Customers





**Figure C-11: Search for Medicine**



**Figure C-12: Medicine's Authenticity Verification**



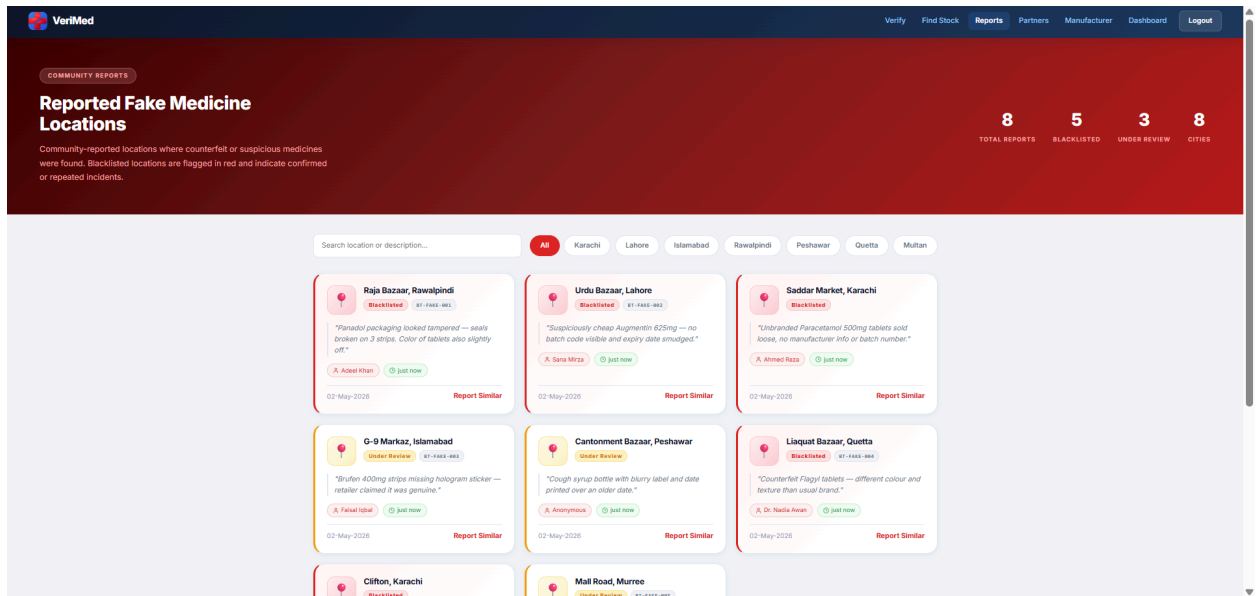


Figure C-15: Reported Fake Medicine Locations

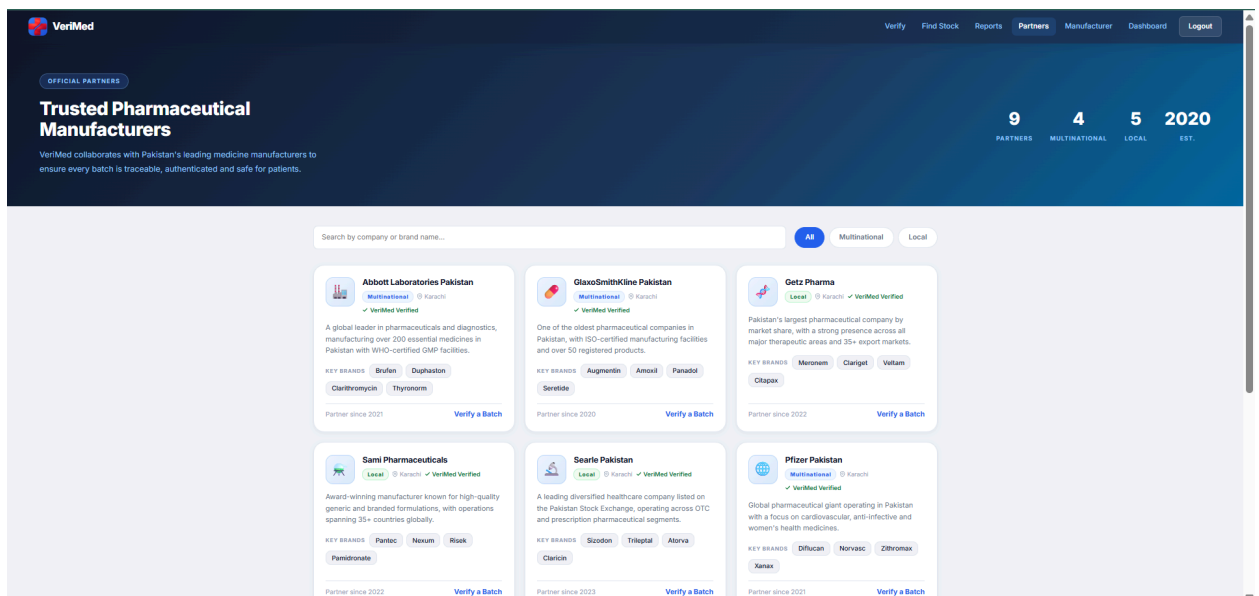
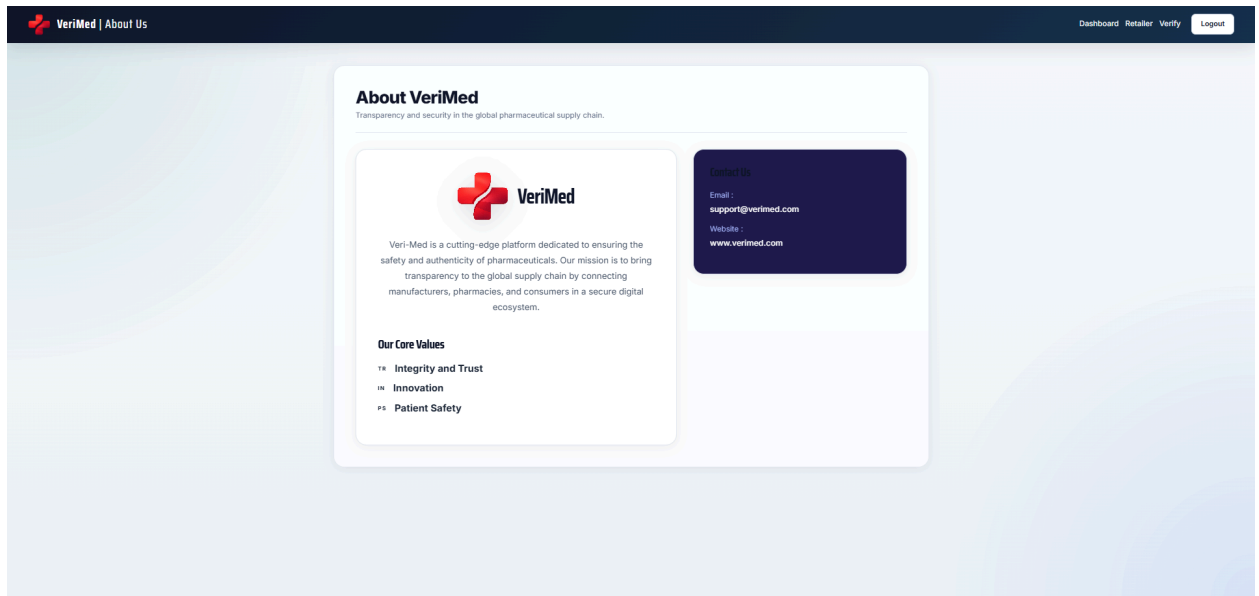
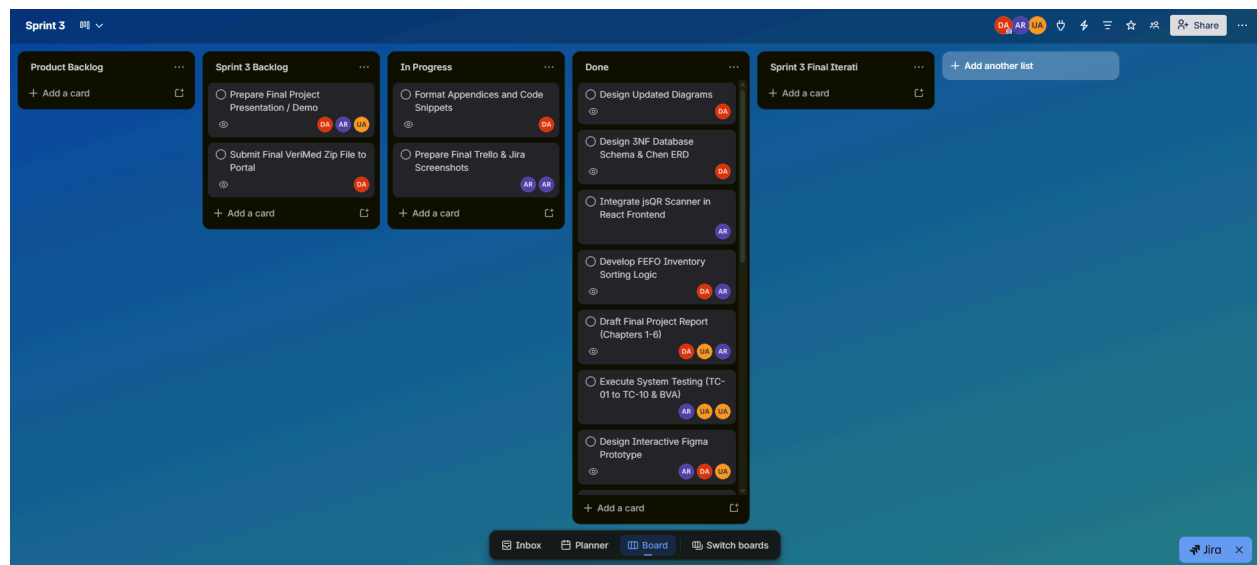


Figure C-16: Trusted Partners



*Figure C-17: About VeriMed*

## Appendix D: Additional Diagrams



*Figure D-1: Trello - Project Backlog and Progress Track*